

51. Graphic LCD Controller (GLCDC)

The multi-function Graphic LCD Controller (GLCDC) supports various color formats and LCD panels. The GLCDC supports the following functions:

- Extended bus master function for accessing graphics data
- Hardware compositing of three graphic screens (single color background, graphic 1, and graphic 2)
- Supports various 32- and 16-bpp (bits per pixel) color formats and 8-, 4-, and 1-bit CLUT (color look-up table) color formats
- Digital RGB interface

Figure 51.1 shows the block diagram of the GLCDC.

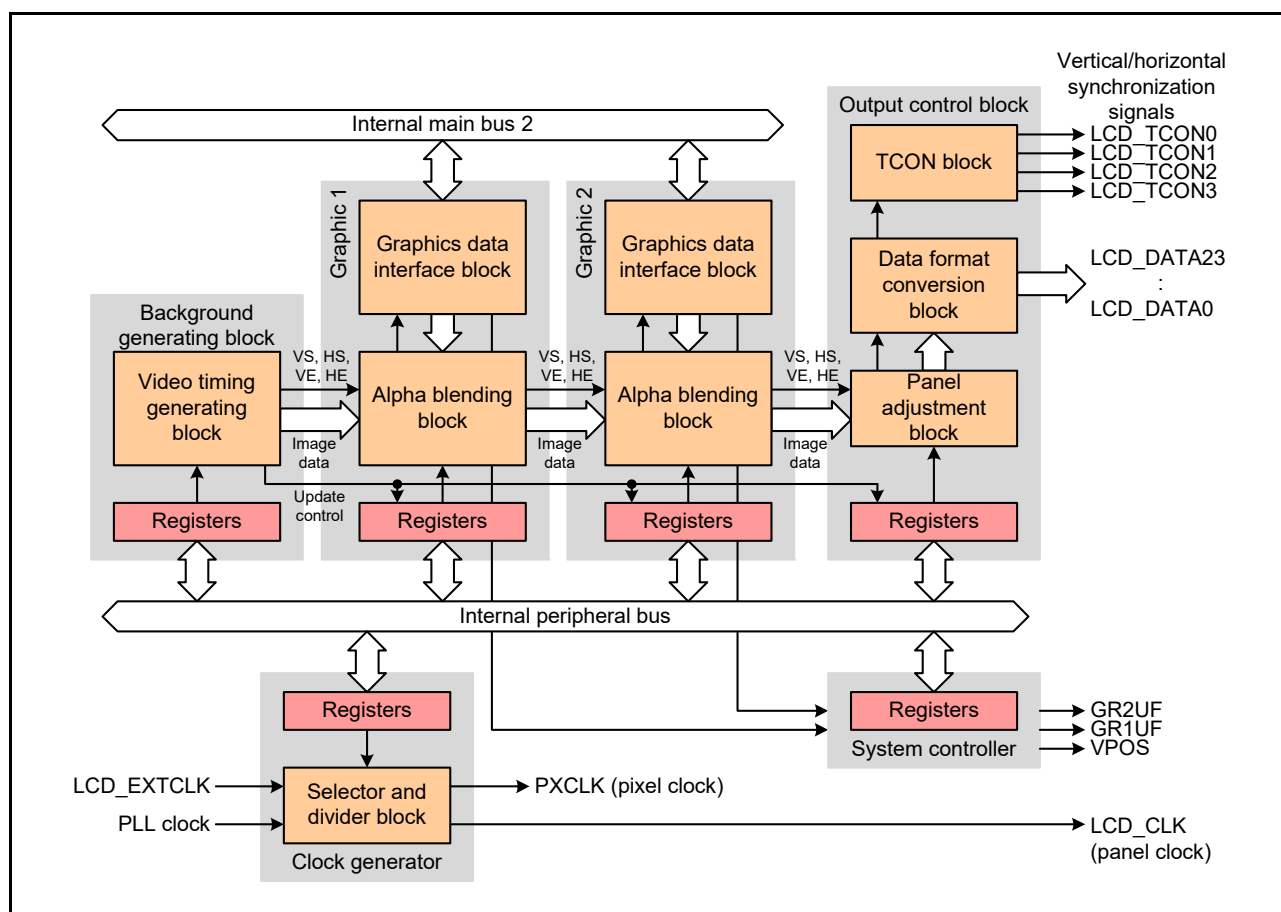


Figure 51.1 GLCDC Block Diagram

51.1 Functional Overview

Table 51.1 lists a functional overview of GLCDC.

Table 51.1 Functional Overview of GLCDC (1/2)

Parameter		Function
Graphics	Graphic screens	Single color background (lowest layer) and two graphic screens Graphic screens can be alpha-blended with the lower-layer screen
	Color format	RGB(-888) progressive format (–: 8 bits, R: 8 bits, G: 8 bits, B: 8 bits; 32 bits in total) ARGB(8888) progressive format (A: 8 bits, R: 8 bits, G: 8 bits, B: 8 bits; 32 bits in total) RGB(565) progressive format (R: 5 bits, G: 6 bits, B: 5 bits; 16 bits in total) ARGB(1555) progressive format (A: CLUT 1 bit, R: 5 bits, G: 5 bits, B: 5 bits; 16 bits in total) ARGB(4444) progressive format (A: 4 bits, R: 4 bits, G: 4 bits, B: 4 bits; 16 bits in total) CLUT(8) progressive format (CLUT index: 8 bits (256 entries)) CLUT(4) progressive format (CLUT index: 4 bits (16 entries)) CLUT(1) progressive format (CLUT index: 1 bit (2 entries))
	CLUT memory	256 words × 32 bits × 2 sets for each graphic screen Data format: ARGB(8888)
	Frame buffer control	The following parameters can be set for the frame buffer: - Base address: Start address of the frame buffer, aligned with a 64-byte boundary - Macro line offset: Offset address between macro lines, aligned with a 64-byte boundary - Number of data transfers: Number of data transfers per macro line - Number of macro lines: Number of macro lines per frame
	Alpha blending	Rectangular alpha blending (blend ratio: 0 to 255) Per-pixel alpha blending (blend ratio: 0 to 255) RGB reference chroma key (replaced with the specified color when the object color matches the preselected color)
Screen format		Total vertical lines: Up to 1024 lines Active vertical lines: 16 to 1020 lines (resolution: 1 line) Vertical front porch: at least 2 lines Vertical back porch: at least 1 line Vertical sync pulse width: fixed to 1 line Total horizontal pixels: Up to 1024 pixels Active horizontal pixels: 16 to 1016 pixels (resolution: 2 pixels) Horizontal front porch: at least 3 pixels Horizontal back porch: at least 1 pixel Horizontal sync pulse width: fixed to 4 pixels Display screen size: from 16 pixels × 16 lines to 1016 pixels × 1020 lines
Data format conversion	Output data format	Four data formats - RGB(888) (parallel 24 bits) - RGB(666) (parallel 18 bits) - RGB(565) (parallel 16 bits) - RGB(888) (serial 8 bits); panel clock period is four times the pixel clock period Programmable bit endian and B/R signal order
	Dither processing	Rounds 10-bit color depth (output from panel adjustment block) to 8-, 6-, or 5-bit color depth (output data format) Supports the following three modes: - Truncate mode - Round-off mode 2×2 pattern dither mode
	Synchronization signal output	Up to four synchronization signals can be output: - Vertical synchronization signal - Horizontal synchronization signal - Vertical display enable/data enable signal - Horizontal display enable/data enable signal

Table 51.1 Functional Overview of GLCDC (2/2)

Parameter	Function	
Panel adjustment processing	Brightness and contrast	10-bit internal processing; the sequence of this processing and gamma correction can be swapped. - Brightness: DC bias adjustment range: from -512 (LSB) to +511 (LSB) - Contrast: gain adjustment range: from 0 to 2 times (from 0/128 to 255/128)
	Gamma correction	10-bit internal processing; the input values are divided into 16 areas and are corrected for linearity in the respective areas. Gain adjustment range in every area: from 0 to 2 times (from 0/1024 to 2047/1024)
Clock generator	Generates the pixel clock (PXCLK) and the panel clock (LCD_CLK). - The maximum frequency of the LCD_CLK is 30 MHz. - The frequency of the PXCLK is the same (parallel RGB) as or 1/4 (serial RGB) of the LCD_CLK. - The maximum frequency of the PXCLK is 1/2 of the PCLKA.	
Interrupts	- Specified line detection interrupt (VPOS) - Graphic 1 underflow interrupt (GR1UF) - Graphic 2 underflow interrupt (GR2UF)	

Table 51.2 GLCDC I/O Pins

Pin name	I/O	Function
LCD_EXTCLK	Input	Panel clock source input pin
LCD_CLK	Output	Panel clock output pin
LCD_DATA23 to LCD_DATA0	Output	LCD signal output pins - RGB(888) signal R/G/B: 8 bits each - RGB(666) signal R/G/B: 6 bits each - RGB(565) signal R/B: 5 bits each, G: 6 bits - Serial RGB signal R/G/B/-: 8 bits each
LCD_TCON3	Output	Synchronization signal (TCON3) output pin
LCD_TCON2	Output	Synchronization signal (TCON2) output pin
LCD_TCON1	Output	Synchronization signal (TCON1) output pin
LCD_TCON0	Output	Synchronization signal (TCON0) output pin

51.1.1 GLCDC Structure

Figure 51.2 shows the structure of the GLCDC.

The GLCDC includes the following blocks:

- Background generating block:
 - Generates the background screen (including the blanking interval), specifies the background color, and generates the synchronization signals for controlling the screens.
- Graphics data interface blocks (2 blocks):
 - Convert the graphics data read from the frame buffer into ARGB(8888) data for internal processing, and perform asynchronous data transfer from peripheral module clock (PCLKA) domain to pixel clock (PXCLK) domain.
- Alpha blending blocks (2 blocks):
 - Combine graphics data with the lower-layer screen and perform alpha blending based on the register settings and the alpha values in the current graphics data.
- TCON block:
 - Generates the vertical and horizontal synchronization and enable signals suited for the specifications of the connected LCD panel, from the internal vertical and horizontal synchronization signals.

- Data format conversion block:
Convert the internal data format into the specific data format suited for the specifications of the LCD panel. Can perform dithering depending on the color depth of the output data format.
- Panel adjustment block:
Adjust brightness and contrast, and performs gamma correction suited for the characteristics of the connected LCD panel. The sequence of gamma correction and brightness/contrast adjustment can be reversed.
- Clock generator block:
Generates the pixel clock (PXCLK) and the panel clock (LCD_CLK) from either LCD_EXTCLK or PLL clock.

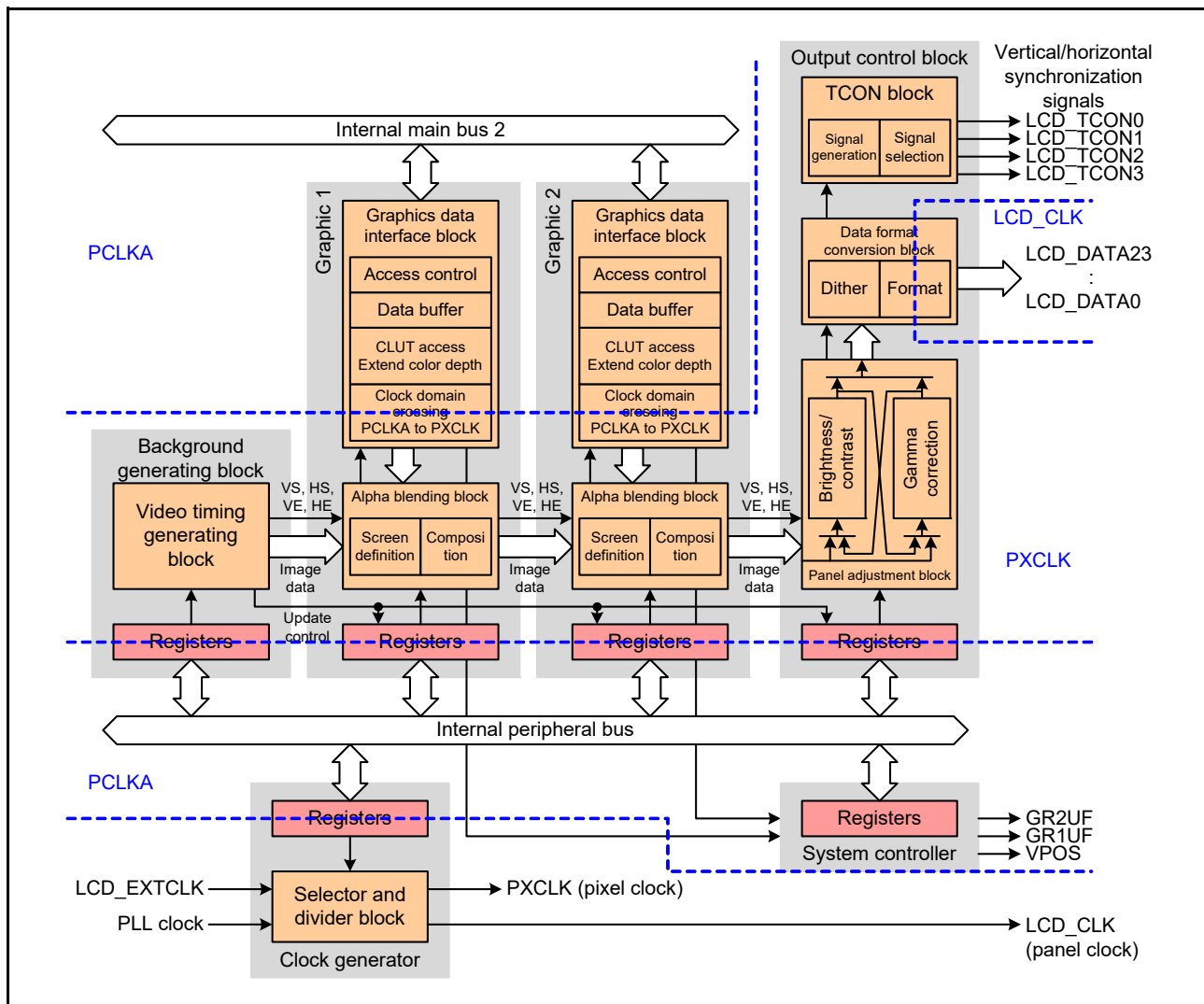


Figure 51.2 GLCDC Structure

51.1.2 Screen Format

Figure 51.3 shows the screen format overview of the GLCDC.

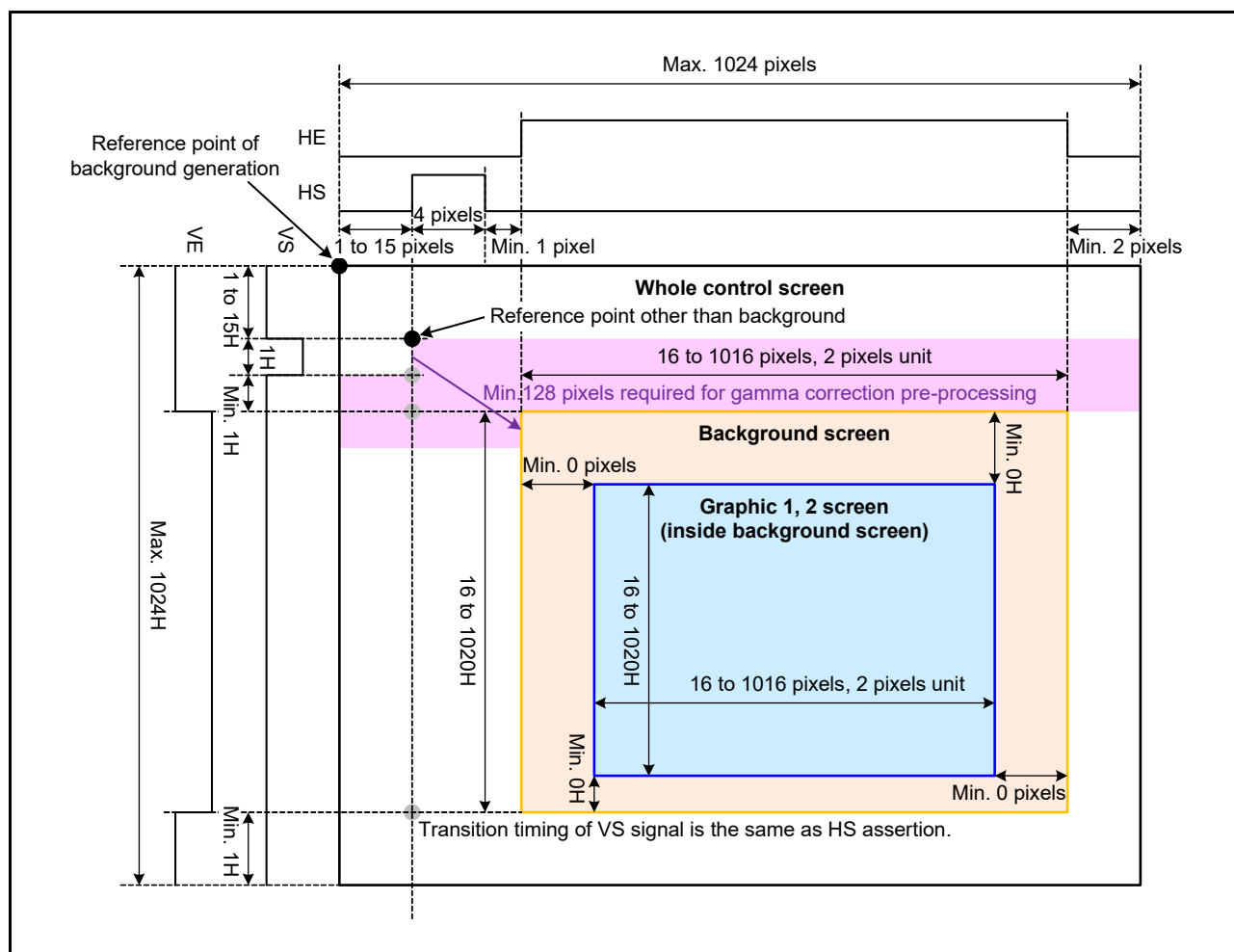


Figure 51.3 Screen Format

The background generating block generates the reference timing signals for operations in GLCDC. The graphics data interface blocks, alpha blending blocks, panel adjustment block, data format conversion block, and TCON block operate based on the synchronization and enable signals, which are sequentially propagated from the background generating block.

51.1.3 Frame Buffer Format

Figure 51.4 shows an example of the frame buffer settings when macro lines are not in use (1 macro line = 1 line).

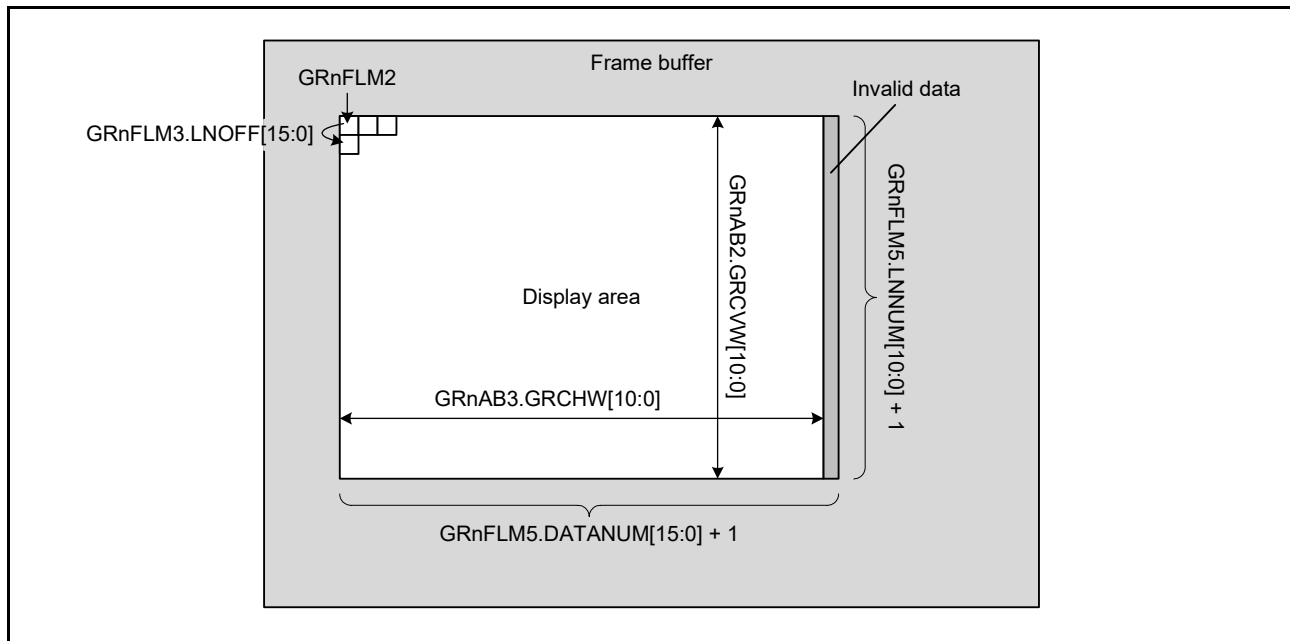


Figure 51.4 Example of Frame Buffer Settings When Macro Lines are not in Use

The GLCDC reads the frame buffer in 64-byte units, so set the GRnFLM2 register and the LNOFF[10:0] bits to multiples of 64.

The width and height of the frame buffer should be set to at least the numbers obtained by $(\text{DATANUM}[15:0] + 1) \times 64$ and $\text{LNNUM}[10:0] + 1$, respectively.

The following is an example of settings when the frame buffer size is 1024 pixels $\times$ 512 lines, the active display size is 480 pixels $\times$ 272 lines, and the color format is CLUT(8).

$$\text{LNOFF}[15:0] = 1024 \times 1 = 1024 = 0400\text{h}$$

$$\text{DATANUM}[15:0] = \text{round}(480 \times 1 \div 64) - 1 = 7 = 0007\text{h}$$

$$\text{LNNUM}[10:0] = 272 - 1 = 271 = 10\text{Fh}$$

$$\text{GRCHW}[10:0] = 480 = 1\text{E}0\text{h}$$

$$\text{GRCVW}[10:0] = 272 = 110\text{h}$$

In this case, data for 512 pixels are read from the frame buffer per line. The data for 32 pixels (invalid data) which are beyond the value set in the GRCHW[10:0] bits are discarded.

Figure 51.5 shows an example of the frame buffer settings when macro lines are in use (1 macro line = multiple lines).

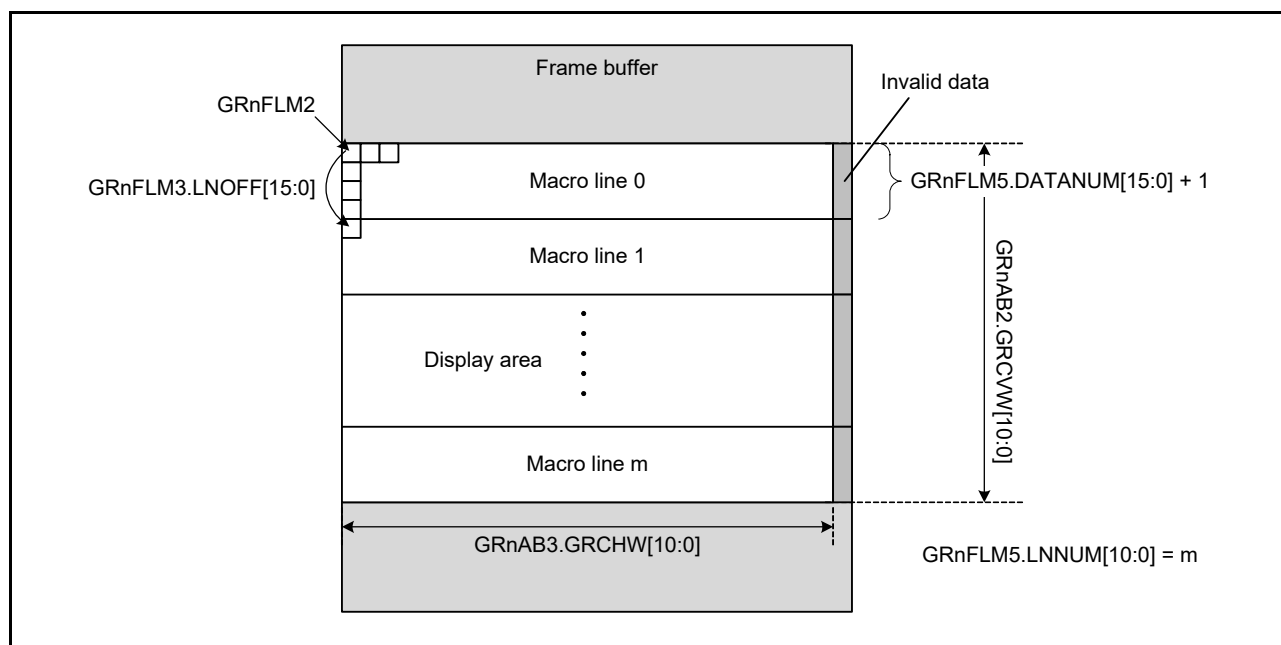


Figure 51.5 Example of Frame Buffer Settings When Macro Lines are in Use

When macro lines are in use, addresses from line to line should be consecutive because there are no registers for specifying the offset between lines. There is no change in terms of the GLCDC reading the frame buffer in 64-byte units and the data which are beyond the value set in the GRCHW[10:0] bits being discarded.

Set the frame buffer width to the smallest value that is a multiple of 64 and greater than or equal to the active display width.

Table 51.3 lists setting examples when the active display size is 480 pixels × 272 lines and the color format is CLUT(8). Set the GRCHW[10:0] bits to 480 (= 1E0h) and the GRCVW[10:0] bits to 272 (= 110h) in every example below.

Table 51.3 Example of Frame Buffer Settings (480 pixels × 272 lines and 8 bpp)

Number of lines per macro line	Number of macro lines per frame	Setting of the LNNUM[10:0] bits	Data size per line	Data size per macro line	Setting of the LNOFF[15:0] bits	Setting of the DATANUM[15:0] bits
1	272	10Fh	512 bytes	512 bytes	Depend on the frame buffer width	0007h
2	136	087h		1024 bytes	0400h	000Fh
4	68	043h		2048 bytes	0800h	001Fh
8	34	021h		4096 bytes	1000h	003Fh
16	17	010h		8192 bytes	2000h	007Fh
17	16	00Fh		8704 bytes	2200h	0087h
34	8	007h		17408 bytes	4400h	010Fh
272	1	000h		139264 bytes	0000h	087Fh

By setting the LNOFF[15:0] bits to 0000h, the same macro line can be repeatedly displayed.

51.1.4 Graphics and Color Look-Up Table (CLUT) Data Formats

The GLCDC has three display screens including a background screen.

Frame data for the background screen is RGB(888) format graphics data stored in the register.

Frame data for the other two screens, graphic 1 and 2, are stored in the memory connected via the internal main bus 2 as 32- or 16-bpp direct color or 8-, 4-, or 1-bpp index color graphics data. Frame data for each screen is read by the graphics data interface block, and extended/converted to ARGB(8888) data for compositing and blending.

Figure 51.6 shows the frame data color formats.

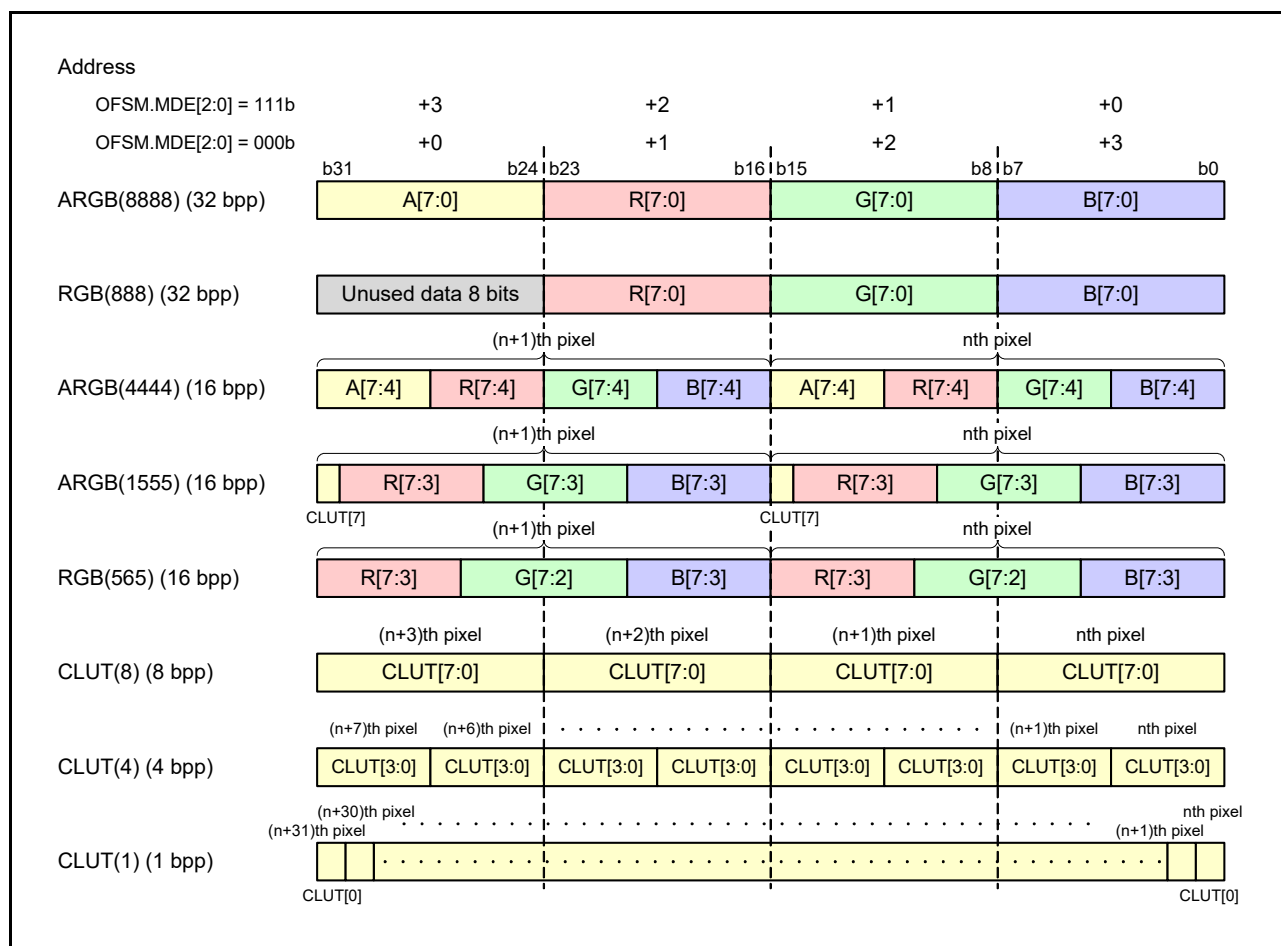


Figure 51.6 Frame Data Color Formats

The A value (alpha value) represents the ratio of combining (blending) pixel data of current graphic screens with the lower-layer graphic screen. The blending process is performed in accordance with the equations given in section 51.1.8, Graphics Data Interfaces. The CLUT value represents the address of the CLUT memory (CLUT 0 or 1 is selected by the registers), and the data in the CLUT memory is ARGB(8888) (32 bpp) format data. Addresses 01h/00h are accessed in CLUT(1) format, 0Fh to 00h in CLUT(4), and FFh to 00h in CLUT(8). Addresses 80h/00h are accessed in ARGB(1555) format.

51.1.5 Output Control (Data Format)

The GLCDC can output data in accordance with the following formats:

- Data formats Parallel: RGB(888), RGB(666), and RGB(565)
 Serial: RGB(888)
 Lower bits are processed in one of the following modes when 10-bit values are rounded to 8-, 6-, or 5- bit values:
 - Truncate mode
 - Round-off mode
 - 2×2 pattern dither processing + truncate mode
- Pixel order R-G-B or B-G-R
- Scan direction for serial RGB(888) Forward scan or reverse scan
- Data output delay of serial RGB(888) 0 to 3 clock cycles
- Pin assignment Little endian or big endian

For details on dither processing, refer to section 51.2.47, Panel Dither Control Register (PANELDTHA).

(1) Bit Assignment of LCD Signals for Parallel RGB(888) Format

Table 51.4 shows RGB signal inputs assigned to the LCD signal outputs for the parallel RGB(888) output format.

Table 51.4 Bit Assignment of RGB Signal Inputs for Parallel RGB(888) Format

Pixel Order	R-G-B		B-G-R	
Pin Assignment	Little Endian	Big Endian	Little Endian	Big Endian
LCD_DATA23	R[7]	R[0]	B[7]	B[0]
LCD_DATA22	R[6]	R[1]	B[6]	B[1]
LCD_DATA21	R[5]	R[2]	B[5]	B[2]
LCD_DATA20	R[4]	R[3]	B[4]	B[3]
LCD_DATA19	R[3]	R[4]	B[3]	B[4]
LCD_DATA18	R[2]	R[5]	B[2]	B[5]
LCD_DATA17	R[1]	R[6]	B[1]	B[6]
LCD_DATA16	R[0]	R[7]	B[0]	B[7]
LCD_DATA15	G[7]	G[0]	G[7]	G[0]
LCD_DATA14	G[6]	G[1]	G[6]	G[1]
LCD_DATA13	G[5]	G[2]	G[5]	G[2]
LCD_DATA12	G[4]	G[3]	G[4]	G[3]
LCD_DATA11	G[3]	G[4]	G[3]	G[4]
LCD_DATA10	G[2]	G[5]	G[2]	G[5]
LCD_DATA9	G[1]	G[6]	G[1]	G[6]
LCD_DATA8	G[0]	G[7]	G[0]	G[7]
LCD_DATA7	B[7]	B[0]	R[7]	R[0]
LCD_DATA6	B[6]	B[1]	R[6]	R[1]
LCD_DATA5	B[5]	B[2]	R[5]	R[2]
LCD_DATA4	B[4]	B[3]	R[4]	R[3]
LCD_DATA3	B[3]	B[4]	R[3]	R[4]
LCD_DATA2	B[2]	B[5]	R[2]	R[5]
LCD_DATA1	B[1]	B[6]	R[1]	R[6]
LCD_DATA0	B[0]	B[7]	R[0]	R[7]

Note: R[7:0], G[7:0], and B[7:0] are RGB pixel data that have been internally processed.

(2) Bit Assignment of LCD Signals for Parallel RGB(666) Format

Table 51.5 shows RGB signal inputs assigned to the LCD signal outputs for the parallel RGB(666) output format.

Table 51.5 Bit Assignment of RGB Signal Inputs for Parallel RGB(666) Format

Pixel Order Pin Assignment	R-G-B		B-G-R	
	Little Endian	Big Endian	Little Endian	Big Endian
LCD_DATA23	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA22	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA21	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA20	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA19	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA18	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA17	R[7]	R[2]	B[7]	B[2]
LCD_DATA16	R[6]	R[3]	B[6]	B[3]
LCD_DATA15	R[5]	R[4]	B[5]	B[4]
LCD_DATA14	R[4]	R[5]	B[4]	B[5]
LCD_DATA13	R[3]	R[6]	B[3]	B[6]
LCD_DATA12	R[2]	R[7]	B[2]	B[7]
LCD_DATA11	G[7]	G[2]	G[7]	G[2]
LCD_DATA10	G[6]	G[3]	G[6]	G[3]
LCD_DATA9	G[5]	G[4]	G[5]	G[4]
LCD_DATA8	G[4]	G[5]	G[4]	G[5]
LCD_DATA7	G[3]	G[6]	G[3]	G[6]
LCD_DATA6	G[2]	G[7]	G[2]	G[7]
LCD_DATA5	B[7]	B[2]	R[7]	R[2]
LCD_DATA4	B[6]	B[3]	R[6]	R[3]
LCD_DATA3	B[5]	B[4]	R[5]	R[4]
LCD_DATA2	B[4]	B[5]	R[4]	R[5]
LCD_DATA1	B[3]	B[6]	R[3]	R[6]
LCD_DATA0	B[2]	B[7]	R[2]	R[7]

Note: R[7:2], G[7:2], and B[7:2] are RGB pixel data that have been internally processed.

(3) Bit Assignment of LCD Signals for Parallel RGB(565) Format

Table 51.6 shows RGB signal inputs assigned to the LCD signal outputs for the parallel RGB(565) output format.

Table 51.6 Bit Assignment of RGB Signal Inputs for Parallel RGB(565) Format

Pixel Order	R-G-B		B-G-R	
Pin Assignment	Little Endian	Big Endian	Little Endian	Big Endian
LCD_DATA23	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA22	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA21	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA20	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA19	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA18	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA17	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA16	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA15	R[7]	R[3]	B[7]	B[3]
LCD_DATA14	R[6]	R[4]	B[6]	B[4]
LCD_DATA13	R[5]	R[5]	B[5]	B[5]
LCD_DATA12	R[4]	R[6]	B[4]	B[6]
LCD_DATA11	R[3]	R[7]	B[3]	B[7]
LCD_DATA10	G[7]	G[2]	G[7]	G[2]
LCD_DATA9	G[6]	G[3]	G[6]	G[3]
LCD_DATA8	G[5]	G[4]	G[5]	G[4]
LCD_DATA7	G[4]	G[5]	G[4]	G[5]
LCD_DATA6	G[3]	G[6]	G[3]	G[6]
LCD_DATA5	G[2]	G[7]	G[2]	G[7]
LCD_DATA4	B[7]	B[3]	R[7]	R[3]
LCD_DATA3	B[6]	B[4]	R[6]	R[4]
LCD_DATA2	B[5]	B[5]	R[5]	R[5]
LCD_DATA1	B[4]	B[6]	R[4]	R[6]
LCD_DATA0	B[3]	B[7]	R[3]	R[7]

Note: R[7:3], G[7:2], and B[7:3] are RGB pixel data that have been internally processed.

(4) Bit Assignment of RGB Signal Inputs for Serial RGB(888) Format

Table 51.7 and Table 51.8 show RGB signal inputs assigned to the LCD signal outputs for the serial RGB(888) output format.

Table 51.7 Bit Assignment of RGB Signal Inputs for Serial RGB(888) Format (1)

Pixel Order	R-G-B															
Pin Assignment	Little Endian								Big Endian							
Scan Direction	Reverse Scan				Forward Scan				Reverse Scan				Forward Scan			
Cycle	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th
LCD_DATA23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
LCD_DATA8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LCD_DATA7	Undefined	B[7]	G[7]	R[7]	R[7]	G[7]	B[7]	Undefined	Undefined	B[0]	G[0]	R[0]	R[0]	G[0]	B[0]	Undefined
LCD_DATA6	Undefined	B[6]	G[6]	R[6]	R[6]	G[6]	B[6]	Undefined	Undefined	B[1]	G[1]	R[1]	R[1]	G[1]	B[1]	Undefined
LCD_DATA5	Undefined	B[5]	G[5]	R[5]	R[5]	G[5]	B[5]	Undefined	Undefined	B[2]	G[2]	R[2]	R[2]	G[2]	B[2]	Undefined
LCD_DATA4	Undefined	B[4]	G[4]	R[4]	R[4]	G[4]	B[4]	Undefined	Undefined	B[3]	G[3]	R[3]	R[3]	G[3]	B[3]	Undefined
LCD_DATA3	Undefined	B[3]	G[3]	R[3]	R[3]	G[3]	B[3]	Undefined	Undefined	B[4]	G[4]	R[4]	R[4]	G[4]	B[4]	Undefined
LCD_DATA2	Undefined	B[2]	G[2]	R[2]	R[2]	G[2]	B[2]	Undefined	Undefined	B[5]	G[5]	R[5]	R[5]	G[5]	B[5]	Undefined
LCD_DATA1	Undefined	B[1]	G[1]	R[1]	R[1]	G[1]	B[1]	Undefined	Undefined	B[6]	G[6]	R[6]	R[6]	G[6]	B[6]	Undefined
LCD_DATA0	Undefined	B[0]	G[0]	R[0]	R[0]	G[0]	B[0]	Undefined	Undefined	B[7]	G[7]	R[7]	R[7]	G[7]	B[7]	Undefined

Note: R[7:0], G[7:0], and B[7:0] are RGB pixel data that have been internally processed.

Table 51.8 Bit Assignment of RGB Signal Inputs for Serial RGB(888) Format (2)

Pixel Order	B-G-R															
Pin Assignment	Little Endian								Big Endian							
Scan Direction	Reverse Scan				Forward Scan				Reverse Scan				Forward Scan			
Cycle	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th
LCD_DATA23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
LCD_DATA8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LCD_DATA7	Undefined	R[7]	G[7]	B[7]	B[7]	G[7]	R[7]	Undefined	Undefined	R[0]	G[0]	B[0]	B[0]	G[0]	R[0]	Undefined
LCD_DATA6	Undefined	R[6]	G[6]	B[6]	B[6]	G[6]	R[6]	Undefined	Undefined	R[1]	G[1]	B[1]	B[1]	G[1]	R[1]	Undefined
LCD_DATA5	Undefined	R[5]	G[5]	B[5]	B[5]	G[5]	R[5]	Undefined	Undefined	R[2]	G[2]	B[2]	B[2]	G[2]	R[2]	Undefined
LCD_DATA4	Undefined	R[4]	G[4]	B[4]	B[4]	G[4]	R[4]	Undefined	Undefined	R[3]	G[3]	B[3]	B[3]	G[3]	R[3]	Undefined
LCD_DATA3	Undefined	R[3]	G[3]	B[3]	B[3]	G[3]	R[3]	Undefined	Undefined	R[4]	G[4]	B[4]	B[4]	G[4]	R[4]	Undefined
LCD_DATA2	Undefined	R[2]	G[2]	B[2]	B[2]	G[2]	R[2]	Undefined	Undefined	R[5]	G[5]	B[5]	B[5]	G[5]	R[5]	Undefined
LCD_DATA1	Undefined	R[1]	G[1]	B[1]	B[1]	G[1]	R[1]	Undefined	Undefined	R[6]	G[6]	B[6]	B[6]	G[6]	R[6]	Undefined
LCD_DATA0	Undefined	R[0]	G[0]	B[0]	B[0]	G[0]	R[0]	Undefined	Undefined	R[7]	G[7]	B[7]	B[7]	G[7]	R[7]	Undefined

Note: R[7:0], G[7:0], and B[7:0] are RGB pixel data that have been internally processed.

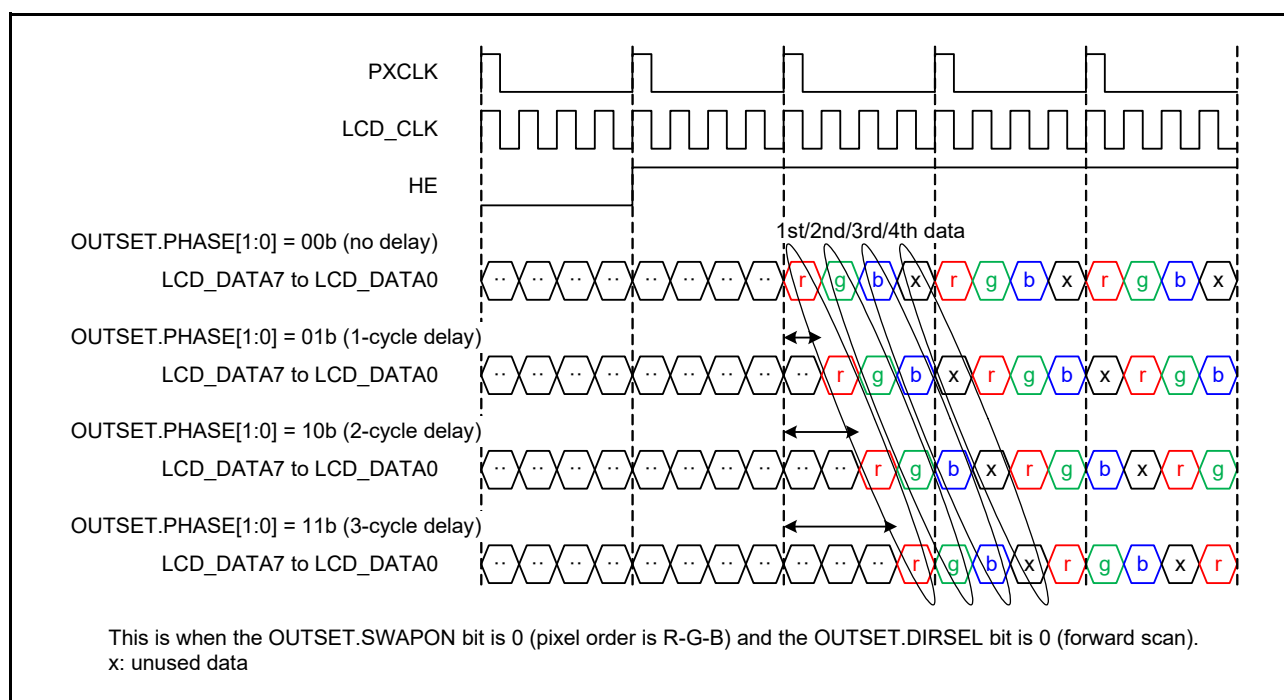


Figure 51.7 Serial RGB(888) (4x Speed) Output Timing

51.1.6 Output Control (Panel Adjustment Process)

The panel adjustment includes the processes of brightness adjustment, contrast adjustment, and RGB gamma correction. Brightness adjustment always precedes contrast adjustment, but RGB gamma correction can either precede or follow brightness and contrast adjustment. In panel adjustment, 10-bit RGB data obtained by extending 8-bit RGB data output from graphic 2 is used, and 10-bit RGB data is also output to the data format conversion block. Figure 51.8 shows the structure of the panel adjustment block.

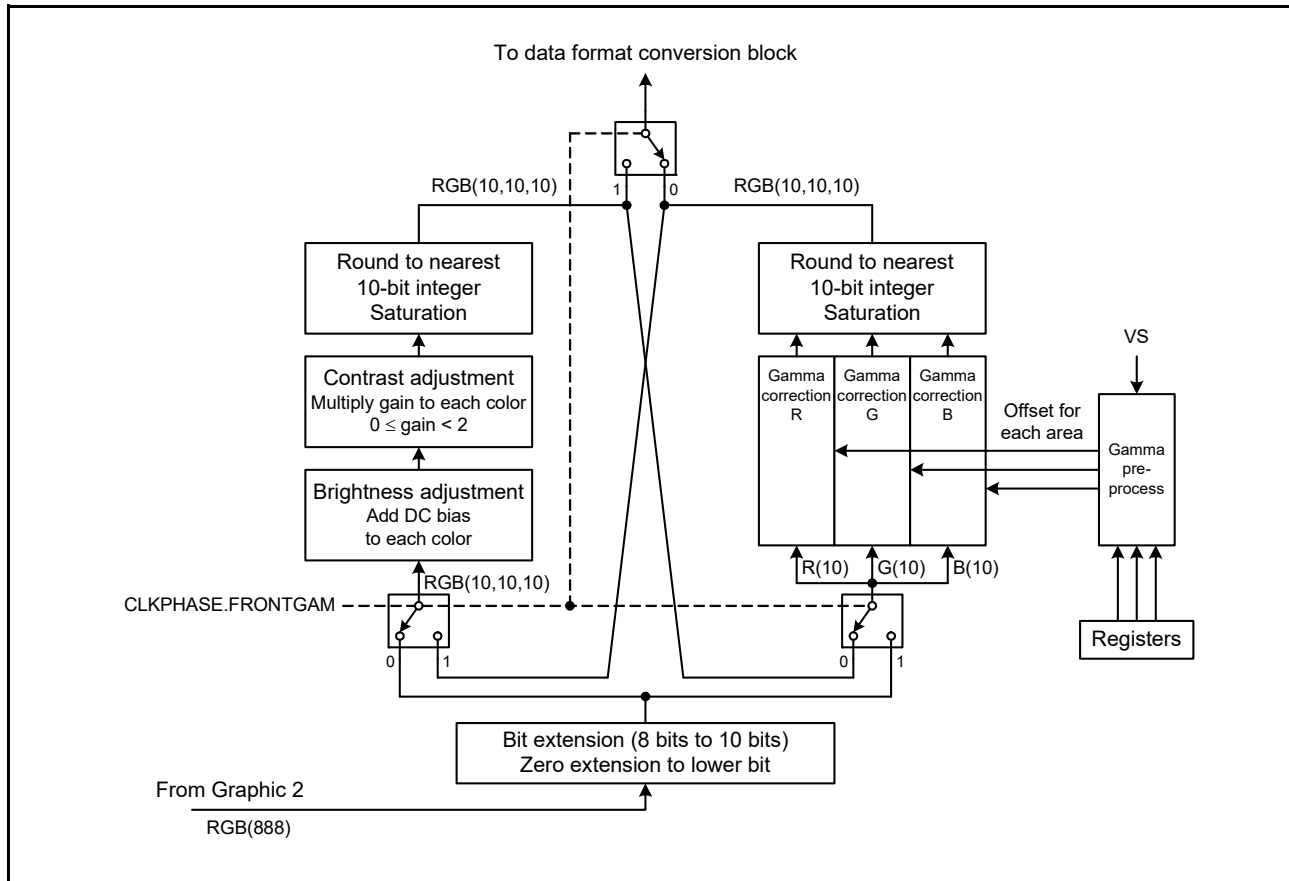


Figure 51.8 Structure of Panel Adjustment Block

51.1.7 Output Control (TCON)

Any of the five signals (STVA, STVB, STHA, STHB, and DE) generated from the internal vertical and horizontal synchronization signals (VS and HS) that have passed through the data format conversion block can be selected for output on four pins: LCD_TCON0, LCD_TCON1, LCD_TCON2, and LCD_TCON3.

The generated signals are completely independent of the image data. They are not affected by the output image format or any internal data process, and no register settings for signal generation affect the output image format or any internal data process. The data enable signal DE, which is to be generated by the TCON block, is the logical AND of the two signals STVB and STHB, which were previously generated by the TCON block. In general, three signals, which includes the signals corresponding to the VS and HS signals plus the DE signal, are used to output the DE signal.

51.1.8 Graphics Data Interfaces

Two circuits (graphic 1 and 2) are provided for reading graphics data, each of which incorporates an access control block, data buffer, CLUT memory, data extension block, and clock domain crossing block.

(1) Access Control Block

After release from the reset state, when the necessary register settings are made and operation of the GLCDC is enabled, this block waits for the input of the VS (vertical synchronization) signal. Every time the VS signal is asserted, the internal states are initialized and access to internal main bus 2 proceeds. Internal main bus 2 is read in 16-beat increments and in 32-bit units (4 bytes $\times$ 16 beats = 64 bytes). All read data are stored in the data buffer, including the invalid data at the end of the line.

(2) Data Buffer

The data buffer is read via a 64-bit read buffer. Reading of the invalid data at the end of the line is skipped. The upper bit in the data stored in the read buffer is used for access to CLUT or data extension, the next data is filled from the data buffer. Filling data to the read buffer takes precedence over writing to the data buffer in case of a conflict.

(3) CLUT Memory

When the pixel data read from the data buffer contains CLUT indices (addresses for accessing to the CLUT memory), the GLCDC accesses to the CLUT memory by reading the corresponding data as the addresses, and stores all read data (in the ARGB(8888) format) or only alpha values (in case of ARGB(1555)) in the buffer of the clock domain crossing block.

Write the data in the CLUT memory via internal peripheral bus.

(4) Data Extension Block

If the pixel data read from the data buffer contains the directly color data, the GLCDC sends the corresponding data to the data extension block. The block extends the data in the ARGB(8888) format and stores them in the buffer of the clock domain crossing block.

When the data contain both of the CLUT indices and direct color data (in case of ARGB(1555)), an A value obtained from the CLUT memory is combined with an RGB value acquired from the data extension block, and is stored in the buffer of the clock domain crossing block.

(5) Clock Domain Crossing Block

This block is a circuit which performs asynchronous transfer of the pixel data (in the ARGB(8888) format) from the peripheral module clock (PCLKA) domain to the pixel clock (PXCLK) domain. The block is composed of a 4-pixel ring buffer. When data for 1 pixel are transferred to the alpha blending block, a sequence of operations consisting of reading from the read buffer, access to CLUT memory, extension of data, combining graphics data, and storage in the ring buffer proceeds with steps included as required, after which 1 pixel data is filled.

51.1.9 Blending

The following three types of processing proceed for the graphics areas specified in the registers. The lower-layer graphics or the background color data of current graphics is output without any processing in the display area outside the graphics area.

- Display screen selection
- RGB reference chroma key
- Alpha blending

(1) Display Screen Selection

Based on the GRnAB1.DISPSEL[1:0] bits settings (n = 1, 2), any of the following data is output to the graphics area:

- Background color: Image data specified in the GRnBASE register.
- Lower-layer graphics: Image data input from the previous block
Output from the background generating block for graphic 1
Output from graphic 1 for graphic 2
- Current graphics: Image data obtained from the graphics data interface block extending the pixel data read by the graphic 1 or 2 block via the internal main bus 2 to ARGB(8888) format data.
- Blending: Image data obtained by blending current graphics data and the lower-layer graphics data based on the alpha values or the register settings.

It is possible to rewrite the GRnAB1 register while display operation is in progress. Setting the BGEN.VEN or GRnVEN.VEN bit to 1 leads to a reflection of the update value to the internal operations when the VS (vertical synchronization) signal is asserted.

(2) RGB Reference Chroma Key

If the RGB value of the ARGB(8888) data input from the graphics data interface block matches the value set in the GRnAB8 register, the ARGB(8888) data is entirely replaced with the value preset in the GRnAB9 register. All the ARGB(8888) data input from the graphics data interface block is subject to this process. If your application excludes CLUT memory output from this process, set the GRnAB7.CKON bit to 0 (disable RGB reference chroma key).

(3) Alpha Blending

If the GRnAB1.DISPSEL[1:0] bits are set to 11b (blending), current graphics and lower-layer graphics are alpha-blended based on the register settings using either of the following two functions.

(a) Rectangular Alpha Blending

The following process is performed for the rectangular area (in the graphics area) preset in the GRnAB4 and GRnAB5 registers:

- (1) Current graphics is blended with the lower-layer graphics according to the A value set in the GRnAB7.ARCDEF[7:0] bits.
- (2) After processing of the number of frames specified in the GRnAB6.ARCRATE[7:0] bits, the value specified in the GRnAB6.ARCCOEF[8:0] bits is added to or subtracted from A value.
- (3) Current graphics is blended with the lower-layer graphics according to this updated A value.
- (4) Repeat (2) and (3). The A value is clipped in the range from 0 to 255.

(b) Per-pixel Alpha Blending

Current graphics is blended with the lower-layer graphics according to the A value of the ARGB(8888) data input from

the graphics data interface block.

Figure 51.9 shows the relation between the graphic display selection and rectangular alpha blending area.

Figure 51.10 shows the update of the alpha value in the rectangular alpha blending area.

Alpha blending is performed based on the following formulas:

- When A value is 255
Rout = Rin1
Gout = Gin1
Bout = Bin1
- When A value is not 255
$$\text{Rout} = (\text{Rin1} \times A + \text{Rin0} \times (256 - A)) / 256$$
$$\text{Gout} = (\text{Gin1} \times A + \text{Gin0} \times (256 - A)) / 256$$
$$\text{Bout} = (\text{Bin1} \times A + \text{Bin0} \times (256 - A)) / 256$$

where,

A: alpha value, which is equivalent to Ain1 for per-pixel alpha blending.

Rin1/Gin1/Bin1: current graphics data

Rin0/Gin0/Bin0: lower-layer graphics data

51.2 Register Descriptions

51.2.1 Graphic n Color Look-up Table m[k] (GRnCLUTm[k]) (n = 1, 2; m = 0, 1; k = 0 to 255)

- Graphic 1 Color Look-up Table 0[k] (GR1CLUT0[k])

Address(es): GLCDC.GR1CLUT0[0] 000E 0000h to GLCDC.GR1CLUT0[255] 000E 03FCh

- Graphic 1 Color Look-up Table 1[k] (GR1CLUT1[k])

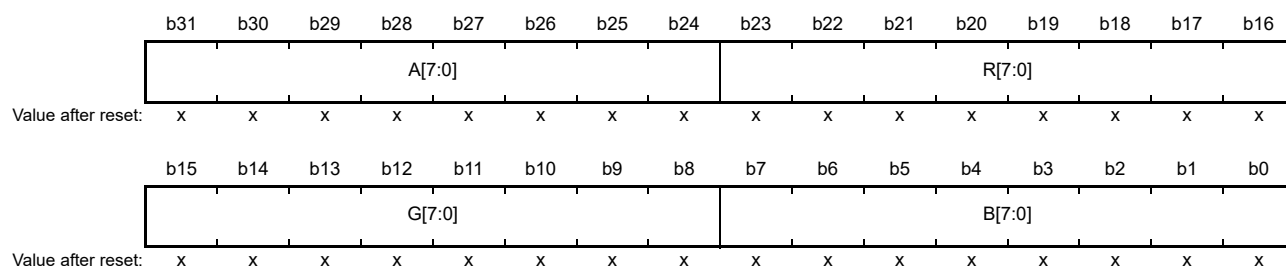
Address(es): GLCDC.GR1CLUT1[0] 000E 0400h to GLCDC.GR1CLUT1[255] 000E 07FCh

- Graphic 2 Color Look-up Table 0[k] (GR2CLUT0[k])

Address(es): GLCDC.GR2CLUT0[0] 000E 0800h to GLCDC.GR2CLUT0[255] 000E 0BFCh

- Graphic 2 Color Look-up Table 1[k] (GR2CLUT1[k])

Address(es): GLCDC.GR2CLUT1[0] 000E 0C00h to GLCDC.GR2CLUT1[255] 000E 0FFCh



x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	B[7:0]	Color Look-up Table B Value Setting	Set the B value for index value k of color look-up table	R/W
b15 to b8	G[7:0]	Color Look-up Table G Value Setting	Set the G value for index value k of color look-up table	R/W
b23 to b16	R[7:0]	Color Look-up Table R Value Setting	Set the R value for index value k of color look-up table	R/W
b31 to b24	A[7:0]	Color Look-up Table A Value Setting	Set the A value for index value k of color look-up table	R/W

The CLUT memory can be rewritten at any time regardless of the setting of the GRnCLUTINT.SEL bit. The updates are reflected to the internal operations directly, not in synchronization with the VS (vertical synchronization) signal. To keep reflection of the CLUT memory contents to the internal operations in synchronization with the VS signal, first rewrite data to the CLUT memory that is not being used for internal operations, and then modify the GRnCLUTINT.SEL bit.

51.2.2 Background Generating Block Operation Control Register (BGEN)

Address(es): GLCDC.BGEN 000E 1000h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SWRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	VEN	—	—	—	—	—	—	—	EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	EN	Background Generating Block Operation Enable	0: Disable background generating block operation 1: Enable background generating block operation	R/W
b7 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b8	VEN	Register Value Reflection Enable	0: Disable the reflection of the register values 1: Enable the reflection of the register values (This bit automatically becomes 0)	R/W
b15 to b9	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b16	SWRST	GLCDC Software Reset Release	0: Execute GLCDC software reset 1: Release GLCDC software reset	R/W
b31 to b17	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

This register controls the GLCDC states. The internal states can be read from the BGMON register and BGEN.VEN. Because the GLCDC, which operates on multiple clocks, requires a certain period for state transition, confirm that the internal state has stabilized when settings are modified. Operation is not guaranteed if settings are modified again before the internal state stabilizes. However, setting the SWRST bit to 0 immediately makes the whole GLCDC reset independently of PXCLK, setting it to 1 releases the reset state.

EN Bit (Background Generating Block Operation Enable)

The EN bit enables or disables the operation of the background generating block. When this bit is set to 0, the operation of the entire GLCDC stops because the reference signal for internal operations generated by the background generating block is no longer being generated.

When setting this bit to 1, set the BGEN.VEN bit to 1 simultaneously to enable register value reflection to internal operations. When setting this bit to 0, set the BGEN.VEN bit to 0 at the same time.

When this bit is changed from 1 to 0, the operation stops at the end of screen generation, unlike the SWRST bit.

When setting this bit to 0, confirm that the BGMON.EN flag has become 0. Operation is not guaranteed if this bit and/or the BGEN.VEN bit are set to 1, or if the settings in other registers are modified before the BGMON.EN flag becomes 0.

VEN Bit (Register Value Reflection Enable)

The VEN bit enables or disables reflection of the GLCDC internal register to the GLCDC internal operation on assertion of the VS (vertical synchronization) signal.

When this bit is set to 1, the signal of GLCDC internal register values reflection control becomes 1 at the immediate start of the background frame, and becomes 0 at the end of the vertical active pixel of the same frame. This internal signal can be monitored with the BGMON.VEN flag. This bit becomes 0 at the same time as the VEN flag.

Set this bit to 1 only when this bit is 0. Operation is not guaranteed if this bit is set to 1 when 1. Also, while this bit is 1, do not modify any register whose value is reflected to the internal operations at the start of the background frame or on assertion of the VS signal. Otherwise, operation is not guaranteed.

SWRST Bit (GLCDC Software Reset Release)

The SWRST bit controls a software reset of the entire GLCDC, not only the background generating block.

When this bit is set to 0, the GLCDC enters the reset state from any operation state. This bit must be set to 1 before the registers are set or operation enabled. The registers can be set while this bit is 1.

Before setting the EN or VEN bit to 1, confirm that PXCLK and LCD_CLK are supplied and that the BGMON.SWRST flag is 1. Operation is not guaranteed if the EN or VEN bit is set to 1 while PXCLK and LCD_CLK are not supplied or the BGMON.SWRST flag is not 1.

51.2.3 Free-Running Period Register (BGPRI)

Address(es): GLCDC.BGPRI 000E 1004h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	FH[10:0]	Horizontal Synchronization Signal Period Setting	017h: 24 pixels : 3FFh: 1024 pixels Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	FV[10:0]	Vertical Synchronization Signal Period Setting	013h: 20 lines (20H) : 3FFh: 1024 lines (1024H) Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

FH[10:0] Bits (Horizontal Synchronization Signal Period Setting)

These bits specify the period of the HS (horizontal synchronization) signal to be generated in the background generating block in pixel units (cycles of the PXCLK). The valid range is 017h to 3FFh. Operation is not guaranteed if a value out of the valid range is set.

FV[10:0] Bits (Vertical Synchronization Signal Period Setting)

These bits specify the period of the VS (vertical synchronization) signal to be generated in the background generating block in line units. The valid range is 013h to 3FFh. Operation is not guaranteed if a value out of the valid range is set.

51.2.4 Synchronization Position Register (BGSYNC)

Address(es): GLCDC.BGSYNC 000E 1008h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	VP[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	HP[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	HP[3:0]	Horizontal Synchronization Signal Assertion Position Setting	0h: Setting prohibited 1h: 1 pixel : Fh: 15 pixels	R/W
b15 to b4	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b19 to b16	VP[3:0]	Vertical Synchronization Signal Assertion Position Setting	0h: Setting prohibited 1h: 1 line (1H) : Fh: 15 lines (15H)	R/W
b31 to b20	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

HP[3:0] Bits (Horizontal Synchronization Signal Assertion Position Setting)

The HP[3:0] bits specify the period before the HS (horizontal synchronization) signal, to be generated in the background generating block, is asserted. The pulse width of the HS signal is fixed to four pixels.

VP[3:0] Bits (Vertical Synchronization Signal Assertion Position Setting)

The VP[3:0] bits specify the period before the VS (vertical synchronization) signal, to be generated in the background generating block, is asserted. The pulse width of the VS signal is fixed to 1H, and the assertion timing is same as the HS signal.

51.2.5 Vertical Size Register (BGVSIZE)

Address(es): GLCDC.BGVSIZE 000E 100Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	VW[10:0]	Vertical Active Display Width Setting	010h: 16 lines (16H) : 3FCh: 1020 lines (1020H) Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	VP[10:0]	Vertical Active Display Start Position Setting	003h: 3 lines (3H) : 3EFh: 1007 lines (1007H) Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

This register specifies the waveform of the VE (vertical data enable) signal generated in the background generating block. The vertical size and position of the display area can be specified.

Specify the vertical active display area between (the assertion position of the VS signal + 2)th line and the (background frame end – 1)th line. Operation is not guaranteed if the area is specified out of the range.

VW[10:0] Bits (Vertical Active Display Width Setting)

The VW[10:0] bits specify the vertical active display lines (high pulse width of the VE signal) in the background screen. The valid range is 010h to 3FCh. Operation is not guaranteed if a value out of the valid range is set. The value of the [BGVSIZE.VP[10:0] + BGVSIZE.VW[10:0] + 1] must be less than or equal to the value of the BGPERI.FV[10:0] bits.

VP[10:0] Bits (Vertical Active Display Start Position Setting)

The VP[10:0] bits specify the vertical active display line start position (the period until the VE signal is asserted) in the background screen. The valid range is 003h to 3EFh. Operation is not guaranteed if a value out of the valid range is set. Set a value greater than or equal to [BGSYNC.VP[3:0] + 2] to these bits.

51.2.6 Horizontal Size Register (BGHSIZE)

Address(es): GLCDC.BGHSIZE 000E 1010h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	HW[10:0]	Horizontal Active Pixel Width Setting	010h: 16 pixels : 3F8h: 1016 pixels Set an even value only in the above range.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	HP[10:0]	Horizontal Active Pixel Start Position Setting	006h: 6 pixels : 3EEh: 1006 pixels Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

This register specifies the waveform of the HE (horizontal data enable) signal generated in the background generating block. The horizontal size and position of the display area can be specified.

Specify the horizontal active display area between (the assertion position of the HS signal + 5)th pixel and (line end – 2)th pixel. Operation is not guaranteed if the area is specified out of the range.

HW[10:0] Bits (Horizontal Active Pixel Width Setting)

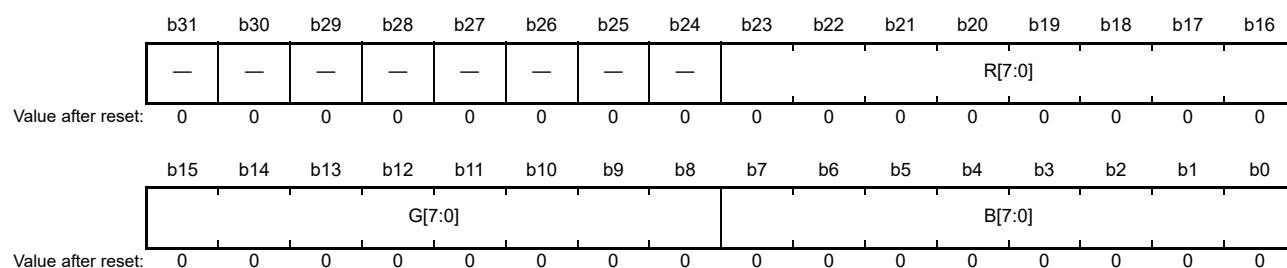
The HW[10:0] bits specify the horizontal active display pixels (high pulse width of the HE signal) in the background screen. The field can be set to any even number from 010h to 3F8h. Operation is not guaranteed if a value out of the valid range is set. The value of the [BGHSIZE.HP[10:0] + BGHSIZE.HW[10:0] + 2] must be less than or equal to the value of the BGPERI.FH[10:0] bits. When serial RGB is selected as the output data format, set the value [the actual horizontal active display pixels + 2] to these bits.

HP[10:0] Bits (Horizontal Active Pixel Start Position Setting)

The HP[10:0] bits specify the horizontal active display pixel start position (the period until the HE signal is asserted) in the background screen. The valid range is 006h to 3EEh. Operation is not guaranteed if a value out of the valid range is set. Set a value greater than or equal to [BGSYNC.HP[3:0] + 5] to these bits.

51.2.7 Background Color Register (BGCOLOR)

Address(es): GLCDC.BGCOLOR 000E 1014h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	B[7:0]	Background Color B Value Setting	Set a value for B channel of the background color to be displayed on the background screen.	R/W
b15 to b8	G[7:0]	Background Color G Value Setting	Set a value for G channel of the background color to be displayed on the background screen.	R/W
b23 to b16	R[7:0]	Background Color R Value Setting	Set a value for R channel of the background color to be displayed on the background screen.	R/W
b31 to b24	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

This register specifies the background color displayed on the active display area (background screen) defined by the BGVSIZ and BGHSIZ registers. Black (000000h) is output to the area other than the active display area (blanking interval).

51.2.8 Background Generating Block Status Monitor Register (BGMON)

Address(es): GLCDC.BGMON 000E 1018h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SWRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	VEN	—	—	—	—	—	—	—	EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	EN	Background Generating Block Operation Monitor Flag	0: Operation of the background generating block is being stopped 1: Operation of the background generating block is in progress	R
b7 to b1	—	Reserved	These bits are read as 0.	R
b8	VEN	Register Value Reflection Enable Monitor Flag	0: The reflection of the register values is disabled 1: The reflection of the register values is enabled	R
b15 to b9	—	Reserved	These bits are read as 0.	R
b16	SWRST	Software Reset Status Monitor Flag	0: GLCDC is in software reset state 1: GLCDC is released from software reset state	R
b31 to b17	—	Reserved	These bits are read as 0.	R

EN Flag (Background Generating Block Operation Monitor Flag)

This flag indicates whether the background generating block is operating or not.

When setting the BGEN.EN bit to 0 (disable background generating block operation), read this flag to confirm that the operation is stopped. Setting the BGEN.EN bit to 0 does not stop the operation until the end of the background frame, unlike when the BGEN.SWRST bit is set to 0.

VEN Flag (Register Value Reflection Enable Monitor Flag)

This flag indicates the status of the signal used for controlling the reflection of the GLCDC internal register values to the internal operations. This signal is asserted at the start of a background frame immediately after setting the BGEN.VEN bit to 1 and negated at the timing of the VE negation.

SWRST Flag (Software Reset Status Monitor Flag)

This flag indicates the software reset state of GLCDC. This bit value indicates the result of the peripheral module clock (PCLKA) resampling the result of the PXCLK sampling the BGEN.SWRST bit. Even if PXCLK is not supplied, setting the BGEN.SWRST bit to 0 sets this flag to 0.

51.2.9 Graphic n Register Update Control Register (GRnVEN) (n = 1, 2)

Address(es): GLCDC.GR1VEN 000E 1100h, GLCDC.GR2VEN 000E 1200h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VEN	Graphic n Register Value Reflection Enable*1	0: Disable reflection of register values 1: Enable reflection of register values (This bit automatically becomes 0)	R/W
b31 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note 1. Do not set this bit to 1 when the BGEN.VEN bit is 1.

VEN Bit (Graphic n Register Value Reflection Enable)

This bit enables or disables reflection of the register values to the graphic n internal operations on assertion of the VS (vertical synchronization) signal.

When the VS signal is asserted immediately after this bit is set to 1, the values set in the GRnFLM2, GRnFLM3, GRnFLM5, GRnFLM6, GRnAB1 to GRnAB9, GRnBASE, and GRnCLUTINT registers are reflected to the internal operations, and then the bit automatically becomes 0. When the BGEN.VEN bit is 1, the register values are reflected to the internal operations on assertion of the VS signal, regardless of the value of this bit.

Set this bit to 1 only when the BGEN.VEN bit and this bit are both 0. Operation is not guaranteed when setting this bit to 1 while one of the bits is 1.

While the BGEN.VEN bit or this bit is 1, do not modify registers GRnFLM2, GRnFLM3, GRnFLM5, GRnFLM6, GRnAB1 to GRnAB9, GRnBASE, and GRnCLUTINT. Otherwise, operation is not guaranteed.

51.2.10 Graphic n Frame Buffer Read Control Register (GRnFLMRD) (n = 1, 2)

Address(es): GLCDC.GR1FLMRD 000E 1104h, GLCDC.GR2FLMRD 000E 1204h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RENB
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RENB	Frame Buffer Read Enable	0: Disable reading of the frame buffer 1: Enable reading of the frame buffer	R/W
b31 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

RENB Bit (Frame Buffer Read Enable)

This bit enables or disables reading of the graphics data on the frame buffer (memory connected via the internal main bus 2).

If the current graphics screen is to be displayed, set this bit to 1 before setting the BGEN.EN bit to 1.

If the current graphics screen is not to be displayed, set this bit to 0 before setting the BGEN.EN bit to 1.

51.2.11 Graphic n Frame Buffer Control Register 2 (GRnFLM2) (n = 1, 2)

Address(es): GLCDC.GR1FLM2 000E 110Ch, GLCDC.GR2FLM2 000E 120Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

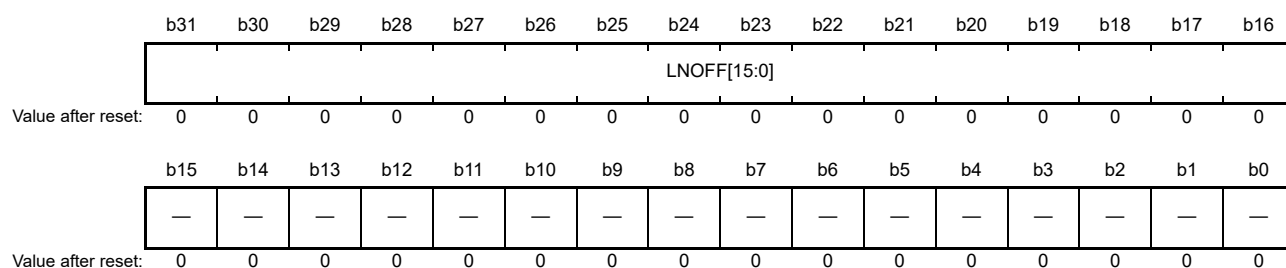
This register specifies the start address of the frame buffer.

Set the lower 6 bits to 0.

This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.12 Graphic n Frame Buffer Control Register 3 (GRnFLM3) (n = 1, 2)

Address(es): GLCDC.GR1FLM3 000E 1110h, GLCDC.GR2FLM3 000E 1210h



Bit	Symbol	Bit Name	Description	R/W
b9 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b10	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b31 to b16	LNOFF[15:0]	Macro Line Offset Setting	Set the offset value between lines (macro lines) on the frame buffer. 16-bit signed integer.	R/W

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

LNOFF[15:0] Bits (Macro Line Offset Setting)

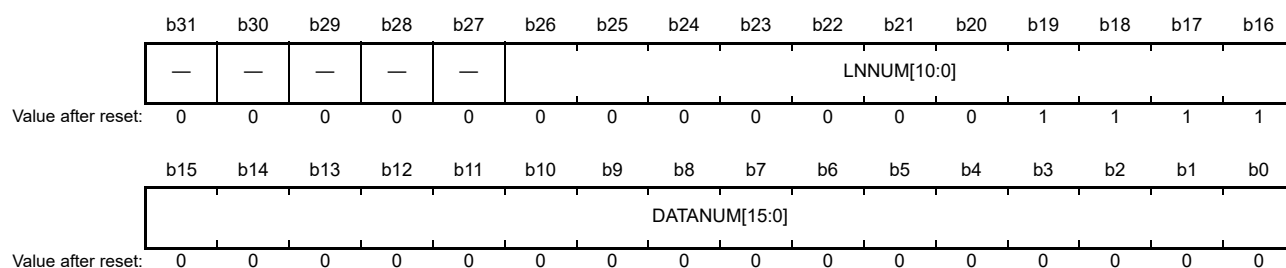
These bits specify the offset from the first address of the line in the frame buffer (macro line) to the first address of the next macro line as a two's complement numbers. Set the lower 6 bits to 0.

The first addresses for each of the macro lines are calculated in the way shown below.

- 1st line: GRnFLM2
- 2nd line: GRnFLM2 + LNOFF[15:0] × 1
- 3rd line: GRnFLM2 + LNOFF[15:0] × 2
- ...
- m-th line: GRnFLM2 + LNOFF[15:0] × (m – 1)

51.2.13 Graphic n Frame Buffer Control Register 5 (GRnFLM5) (n = 1, 2)

Address(es): GLCDC.GR1FLM5 000E 1118h, GLCDC.GR2FLM5 000E 1218h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	DATANUM[15:0]	Single Line Data Transfer Count Setting	0000h: Once (64 bytes) : FFFFh: 65536 times (4 Mbytes)	R/W
b26 to b16	LNNUM[10:0]	Single Frame Line Count Setting	000h: 1 macro line : 3FBh: 1020 macro lines Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

DATANUM[15:0] Bits (Single Line Data Transfer Count Setting)

These bits specify the number of data transfers for one line in the frame buffer (one macro line). 64 bytes (4 bytes (32 bits) × 16 words) of data can be read in each single transfer. If the number of bytes per macro line cannot be divided by 64, set these bits for the nearest integer produced by rounding up.

When the number of data transfers set in these bits is completed, the internal address pointer is overwritten by the first address of the next macro line.

LNNUM[10:0] Bits (Single Frame Line Count Setting)

These bits specify the number of macro lines on the frame buffer.

When graphics data for the number of macro lines set in these bits is read, the frame end signal is output and the value of the GRnFLM2 register is reloaded to the internal address pointer.

The followings are two setting examples of this register. In these examples, the active display size is 480 pixels × 272 lines and the color depth is 16 bpp.

Ex. 1) One macro line is configured of 480 pixels.

$$\text{DATANUM}[15:0] = 480 \times 2 \div 64 - 1 = 14 = 000\text{Eh}$$

$$\text{LNNUM}[10:0] = 272 - 1 = 271 = 10\text{Fh}$$

Ex. 2) One macro line is configured of 16 scanlines (480 × 16 pixels).

$$\text{DATANUM}[15:0] = 480 \times 2 \times 16 \div 64 - 1 = 239 = 00\text{EFh}$$

$$\text{LNNUM}[10:0] = 272 \div 16 - 1 = 16 = 010\text{h}$$

51.2.14 Graphic n Frame Buffer Control Register 6 (GRnFLM6) (n = 1, 2)

Address(es): GLCDC.GR1FLM6 000E 111Ch, GLCDC.GR2FLM6 000E 121Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	FORMAT[2:0]			—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b27 to b0	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b30 to b28	FORMAT[2:0]	Frame Buffer Color Format Setting	b30 b28 0 0 0: RGB(565) (16 bpp) 0 0 1: RGB(-888) (32 bpp, the upper 8 bits are invalid) 0 1 0: ARGB(1555) (16 bpp, 1 bit of A is CLUT index) 0 1 1: ARGB(4444) (16 bpp) 1 0 0: ARGB(8888) (32 bpp) 1 0 1: CLUT(8) (8 bpp) 1 1 0: CLUT(4) (4 bpp) 1 1 1: CLUT(1) (1 bpp)	R/W
b31	—	Reserved	This bit is read as 0. Writing to this bit has no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

FORMAT[2:0] Bits (Frame Buffer Color Format Setting)

These bits specify the color format of the graphics data stored in the corresponding frame buffer.

The CLUT indices 1 or 0, Fh to 0h, and FFh to 00h are stored in CLUT(1), CLUT(4), and CLUT(8) respectively, and used as addresses in CLUT memory access (01h or 00h, 0Fh to 00h, and FFh to 00h respectively).

For ARGB(1555), RGB data (5 bits per color channel) are stored in the 15 bits following the CLUT index with the upper bit 1 or 0 (the CLUT address is 80h or 00h).

For ARGB(8888) and ARGB(4444), the alpha values are stored in the upper 8 or 4 bits and RGB values are stored in the other bits.

RGB(-888) and RGB(565) contain RGB data only.

51.2.15 Graphic n Alpha Blending Control Register 1 (GRnAB1) (n = 1, 2)

Address(es): GLCDC.GR1AB1 000E 1120h, GLCDC.GR2AB1 000E 1220h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	ARCON	—	—	—	ARCDISPON	—	—	—	GRCDISPON	—	—	DISPSEL[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	DISPSEL[1:0]	Display Screen Control	b1 b0 0 0: Displays the background color (value set in the GRnBASE register) 0 1: Displays lower-layer graphics 1 0: Displays current graphics 1 1: Blended display of current graphics with lower-layer graphics	R/W
b3, b2	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b4	GRCDISPON	Graphics Area Frame Display Control	0: The frame of the graphics area is not displayed 1: The frame of the graphics area is displayed	R/W
b7 to b5	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b8	ARCDISPON	Rectangular Alpha Blending Area Frame Display Control	0: The frame of the rectangular alpha blending area is not displayed 1: The frame of the rectangular alpha blending area is displayed	R/W
b11 to b9	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b12	ARCON	Alpha Blending Control	0: Per-pixel alpha blending 1: Rectangular alpha blending	R/W
b31 to b13	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

DISPSEL[1:0] Bits (Display Screen Control)

These bits control the graphic display screen. When the bits are 00b (background color), the background color specified in the GRnBASE register is displayed. When the bits are 01b (lower-layer graphics), the image data input from the previous block is displayed (the background screen for graphic 1, and the output from graphic 1 for graphic 2). When the bits are 10b (current graphics), RGB(888) in data in the ARGB(8888) format, which is generated in the graphics data interface block, is displayed. Table 51.9 lists the relations between the register settings and display on the screen and Figure 51.9 shows the relations among the areas.

GRCDISPON Bit (Graphics Area Frame Display Control)

This bit specifies whether or not the given frame is displayed in the graphic area. When the bit is 1 (displayed), the frame in the graphics area is enclosed by the lines in the color FFh (for each of R, G, and B). The lines around the outer boundaries are 1-pixel wide.

ARCDISPON Bit (Rectangular Alpha Blending Area Frame Display Control)

This bit specifies whether or not the given frame is displayed in the rectangular alpha blending area. When this bit is 1 (displayed), the frame in the rectangular alpha blending area is enclosed by the lines in the color FFh (for each of R, G,

and B). The lines around the outer boundaries are 1-pixel wide.

ARCON Bit (Alpha Blending Control)

This bit controls the alpha blending method in a rectangular alpha blending area. When this bit is 1 (rectangular alpha blending), alpha blending is processed in accordance with the alpha value for the rectangular alpha blending, not in accordance with the alpha value for each pixel generated by the graphics data interface block. In the areas outside the rectangular alpha blending area within the graphics area, alpha blending is processed in accordance with the alpha value for each pixel generated by the graphics data interface block.

Table 51.9 Display Screen Selections (n = 1, 2)

GRnAB1.DISPSEL[1:0] bits	GRnAB1.ARCON bit	GRnAB7.CKON bit	Within Rectangular Alpha Blending Area	Outside Rectangular Alpha Blending Area, but within Graphics Area	Outside Graphics Area, but within Display Area
00b (back ground color)	0 (per-pixel alpha blending)	0 (no chroma keying)	— (no areas)	Background color	Background color
01b (lower-layer graphics)	0 (per-pixel alpha blending)	0 (no chroma keying)	— (no areas)	Lower-layer graphics	Lower-layer graphics
10b (current graphics)	0 (per-pixel alpha blending)	0 (no chroma keying)	— (no areas)	Current graphics	Background color
11b (blended display)	0 (per-pixel alpha blending)	0 (no chroma keying)	— (no areas)	Per-pixel alpha blending of current graphics and lower-layer graphics	Lower-layer graphics
11b (blended display)	0 (per-pixel alpha blending)	1 (chroma keying)	— (no areas)	Per-pixel alpha blending of chroma keyed current graphics and lower-layer graphics	Lower-layer graphics
11b (blended display)	1 (rectangular alpha blending)	0 (no chroma keying)	Rectangular alpha blending of current graphics and lower-layer graphics	Per-pixel alpha blending of current graphics and lower-layer graphics	Lower-layer graphics
11b (blended display)	1 (rectangular alpha blending)	1 (chroma keying)	Rectangular alpha blending of current graphics and lower-layer graphics	Per-pixel alpha blending of chroma keyed current graphics and lower-layer graphics	Lower-layer graphics

Note: Operation is not guaranteed when any other value is set.

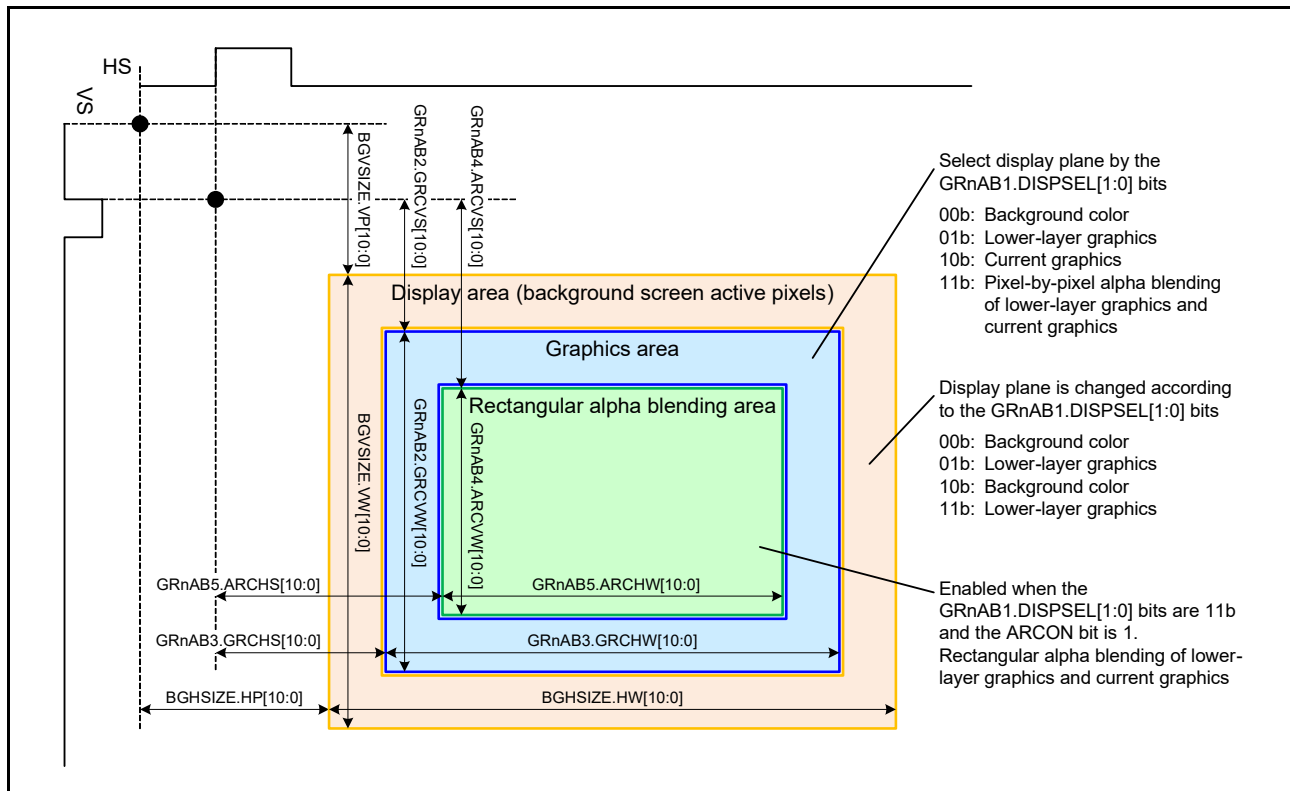


Figure 51.9 Selection of Graphic Display Screen (n = 1, 2)

51.2.17 Graphic n Alpha Blending Control Register 3 (GRnAB3) (n = 1, 2)

Address(es): GLCDC.GR1AB3 000E 1128h, GLCDC.GR2AB3 000E 1228h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GRCHW[10:0]	Graphics Area Horizontal Width Setting	010h: 16 pixels : 3F8h: 1016 pixels Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GRCHS[10:0]	Graphics Area Horizontal Start Position Setting	005h: 5 pixels : 3EDh: 1005 pixels Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

GRCHW[10:0] Bits (Graphics Area Horizontal Width Setting)

These bits specify the horizontal active pixels of the graphics area. Set an even value to the bits.

GRCHS[10:0] Bits (Graphics Area Horizontal Start Position Setting)

These bits specify the horizontal start position of the active pixels of the graphics area. The reference point is the assertion of the HS (horizontal synchronization) signal.

For the relation between the register settings and the graphics area size, refer to Figure 51.9.

51.2.18 Graphic n Alpha Blending Control Register 4 (GRnAB4) (n = 1, 2)

Address(es): GLCDC.GR1AB4 000E 112Ch, GLCDC.GR2AB4 000E 122Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	ARCVW[10:0]	Rectangular Alpha Blending Area Vertical Width Setting	001h: 1 line (1H) : 3FCh: 1020 lines (1020H) Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	ARCVS[10:0]	Rectangular Alpha Blending Area Vertical Start Position Setting	002h: 2 lines (2H) : 3EEh: 1006 lines (1006H) Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

ARCVW[10:0] Bits (Rectangular Alpha Blending Area Vertical Width Setting)

These bits specify the vertical active scanlines of the rectangular alpha blending area.

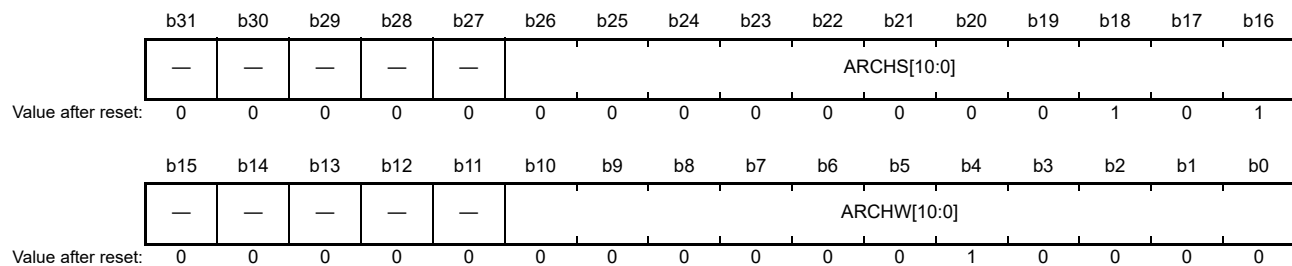
ARCVS[10:0] Bits (Rectangular Alpha Blending Area Vertical Start Position Setting)

These bits specify the vertical start position of the active scanlines of the rectangular alpha blending area. The reference point is the assertion of the VS (vertical synchronization) signal.

For the relation between the register settings and the graphics area size, refer to Figure 51.9.

51.2.19 Graphic n Alpha Blending Control Register 5 (GRnAB5) (n = 1, 2)

Address(es): GLCDC.GR1AB5 000E 1130h, GLCDC.GR2AB5 000E 1230h



Bit	Symbol	Bit Name	Description	R/W
b10 to b0	ARCHW[10:0]	Rectangular Alpha Blending Area Horizontal Width Setting	001h: 1 pixel : 3F8h: 1016 pixels Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	ARCHS[10:0]	Rectangular Alpha Blending Area Horizontal Start Position Setting	005h: 5 pixels : 3EDh: 1005 pixels Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

ARCHW[10:0] Bits (Rectangular Alpha Blending Area Horizontal Width Setting)

These bits specify the horizontal active pixels of the rectangular alpha blending area.

ARCHS[10:0] Bits (Rectangular Alpha Blending Area Horizontal Start Position Setting)

These bits specify the horizontal start position of the active pixels of the rectangular alpha blending area. The reference point is the assertion of the HS (horizontal synchronization) signal.

For the relation between the register settings and the graphics area size, refer to Figure 51.9.

51.2.20 Graphic n Alpha Blending Control Register 6 (GRnAB6) (n = 1, 2)

Address(es): GLCDC.GR1AB6 000E 1134h, GLCDC.GR2AB6 000E 1234h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	ARCCOEF[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	ARCRATE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	ARCRATE[7:0]	Alpha Blending Updating Rate Setting	00h: 1 frame : FFh: 256 frames	R/W
b15 to b8	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b24	ARCCOEF[8]	Alpha Coefficient Setting	This bit specifies the sign, i.e. whether the ARCCOEF[7:0] bits indicate an increase or decrease in the alpha value. 0: Add 1: Subtract	R/W
b23 to b16	ARCCOEF[7:0]		These bits specify the absolute value of the change in the alpha value. The valid range is from 0 to 255.	R/W
b31 to b25	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

ARCRATE[7:0] Bits (Alpha Blending Updating Rate Setting)

These bits specify the updating rate of alpha value for rectangular alpha blending in frame units.

ARCCOEF[8:0] Bits (Alpha Coefficient Setting)

These bits specify the alpha coefficient for rectangular alpha blending.

The ARCCOEF[8] bit represents the sign and the ARCCOEF[7:0] bits represent the value added to or subtracted by.

In rectangular alpha blending, current graphics are faded in or out with the GRnAB7.ARCDEF[7:0], GRnAB6.ARCCOEF[8:0], and GRnAB6.ARCRATE[7:0] bits. When the GRnAB1.ARCON bit is set to 1 after setting the initial alpha value in the GRnAB7.ARCDEF[7:0] bits, the value in GRnAB6.ARCCOEF[8:0] is added to or subtracted from the alpha value each time the VS signal is asserted the number of times set in the GRnAB6.ARCRATE[7:0] bits. Figure 51.10 shows change of the alpha value.

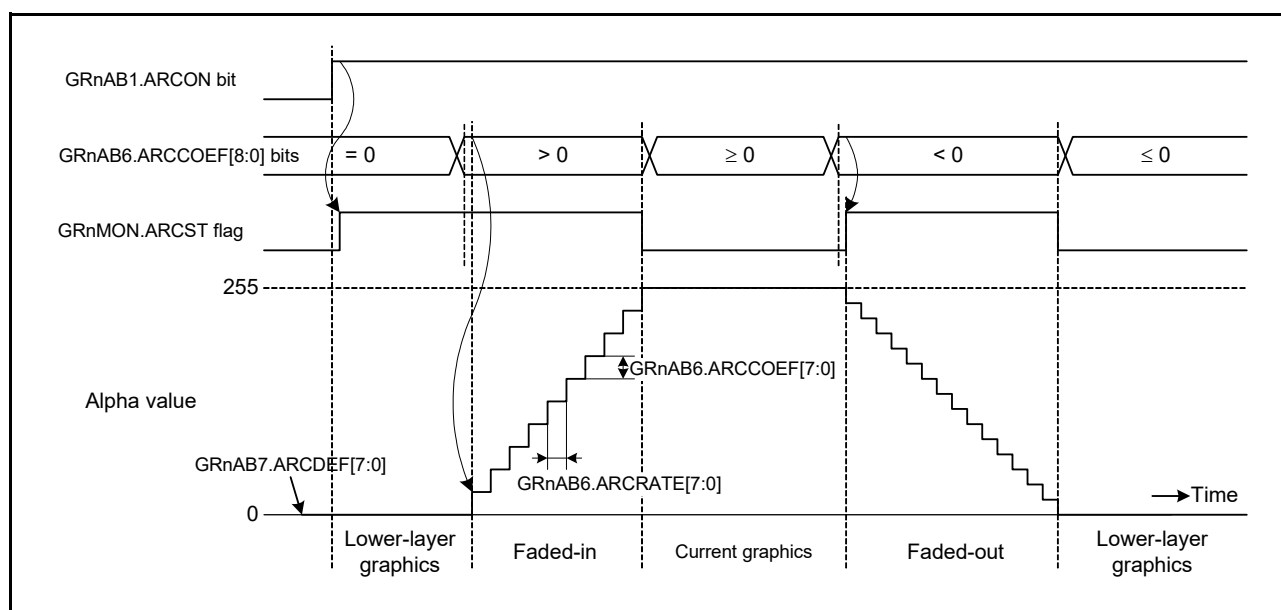


Figure 51.10 Updating of Alpha Value (n = 1, 2)

51.2.21 Graphic n Alpha Blending Control Register 7 (GRnAB7) (n = 1, 2)

Address(es): GLCDC.GR1AB7 000E 1138h, GLCDC.GR2AB7 000E 1238h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	ARCDEF[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CKON
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CKON	Chroma Key Compositing Control	0: Disable RGB reference chroma key compositing 1: Enable RGB reference chroma key compositing	R/W
b15 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b23 to b16	ARCDEF[7:0]	Initial Alpha Value Setting	Set the initial alpha value for rectangular alpha blending	R/W
b31 to b24	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

CKON Bit (Chroma Key Compositing Control)

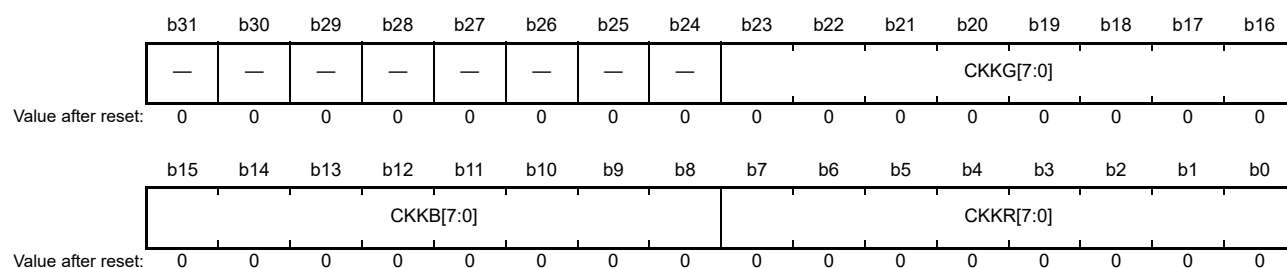
This bit enables or disables RGB reference chroma key compositing. This function is enabled only for the area that is outside the alpha blending area and within the graphics area, when this bit is 1 and the GRnAB1.DISPSEL[1:0] bits are 11b (blended display). For details, refer to Table 51.9.

ARCDEF[7:0] Bits (Initial Alpha Value Setting)

These bits specify the initial alpha value for rectangular alpha blending. For changes in the alpha value during fade-in or fade-out processing of the current graphics by using these bits, refer to Figure 51.10.

51.2.22 Graphic n Alpha Blending Control Register 8 (GRnAB8) (n = 1, 2)

Address(es): GLCDC.GR1AB8 000E 113Ch, GLCDC.GR2AB8 000E 123Ch



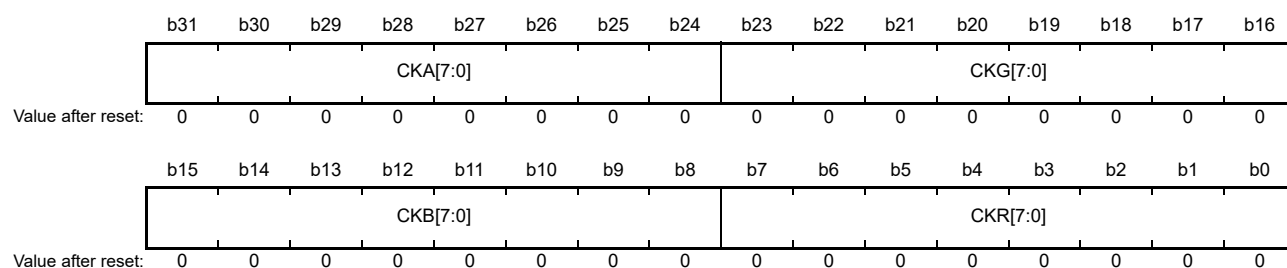
Bit	Symbol	Bit Name	Description	R/W
b7 to b0	CKKR[7:0]	Chroma Key Compositing Target R Value Setting	Set the R value to be compared with the R value of the current graphics.	R/W
b15 to b8	CKKB[7:0]	Chroma Key Compositing Target B Value Setting	Set the B value to be compared with the B value of the current graphics.	R/W
b23 to b16	CKKG[7:0]	Chroma Key Compositing Target G Value Setting	Set the G value to be compared with the G value of the current graphics.	R/W
b31 to b24	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

When the RGB values in the pixel data (ARGB(8888)) of the current graphics match the values set in this register while the GRnAB7.CKON bit is 1 (RGB reference chroma key compositing is enabled), the pixel data is replaced by the values set in the GRnAB9 register. In per-pixel alpha blending at later stages, the replaced alpha value is used.

51.2.23 Graphic n Alpha Blending Control Register 9 (GRnAB9) (n = 1, 2)

Address(es): GLCDC.GR1AB9 000E 1140h, GLCDC.GR2AB9 000E 1240h



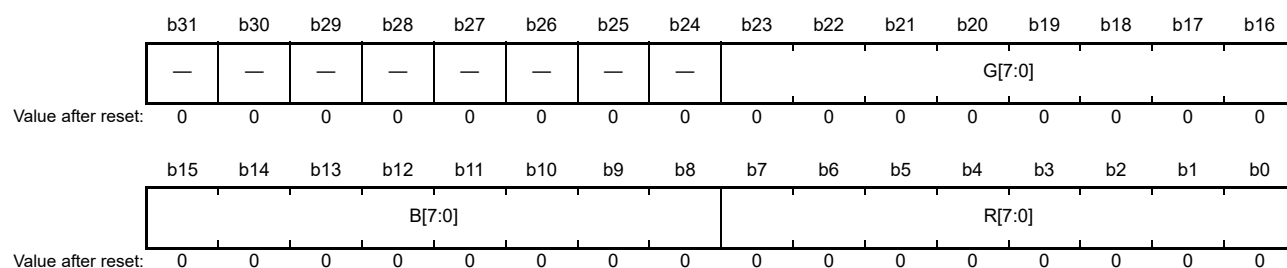
Bit	Symbol	Bit Name	Description	R/W
b7 to b0	CKR[7:0]	Chroma Key Compositing Replacing R Value Setting	Specify an R value after replacement in RGB reference chroma key compositing.	R/W
b15 to b8	CKB[7:0]	Chroma Key Compositing Replacing B Value Setting	Specify a B value after replacement in RGB reference chroma key compositing.	R/W
b23 to b16	CKG[7:0]	Chroma Key Compositing Replacing G Value Setting	Specify a G value after replacement in RGB reference chroma key compositing.	R/W
b31 to b24	CKA[7:0]	Chroma Key Compositing Replacing A Value Setting	Specify an A value after replacement in RGB reference chroma key compositing.	R/W

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

When the pixel data is to be replaced (an RGB value of the current graphics matches with the value set in the GRnAB8 register) while the GRnAB7.CKON bit is 1 (RGB reference chroma key compositing is enabled), the pixel data is replaced by the set value in this register. In per-pixel alpha blending at later stages, the values set in this register is used.

51.2.24 Graphic n Background Color Control Register (GRnBASE) (n = 1, 2)

Address(es): GLCDC.GR1BASE 000E 114Ch, GLCDC.GR2BASE 000E 124Ch



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	R[7:0]	Background Color R Value Setting	Set the R value of the Background color.	R/W
b15 to b8	B[7:0]	Background Color B Value Setting	Set the B value of the Background color.	R/W
b23 to b16	G[7:0]	Background Color G Value Setting	Set the G value of the Background color.	R/W
b31 to b24	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

When the GRnAB1.DISPSEL[1:0] bits are set to 00b (background color), the RGB data set in this register is output to the entire display area. When the GRnAB1.DISPSEL[1:0] bits are set to 10b (current graphics), the RGB data is output to the outside the graphics area within the display area.

51.2.25 Graphic n CLUT/Interrupt Control Register (GRnCLUTINT) (n = 1, 2)

Address(es): GLCDC.GR1CLUTINT 000E 1150h, GLCDC.GR2CLUTINT 000E 1250h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	LINE[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	LINE[10:0]	Detecting Scanline Setting*1	000h: 1st line : 3FFh: 1024th line Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b16	SEL	CLUT Control	0: Select Color Look-up Table 0 1: Select Color Look-up Table 1	R/W
b31 to b17	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GRnVEN.VEN bit or the BGEN.VEN bit is 1.

Note 1. These bits are only effective for graphic 2.

LINE[10:0] Bits (Detecting Scanline Setting)

These bits specify the scanline which generates a VPOS interrupt. When operation reaches the scanline specified in these bits, the detection signal is informed to the system control block on the HS signal assertion. To catch the informed detection signal and request the VPOS interrupt, set the DTCTEN.VPOSDTC and INTEN.VPOSINTEN bits to 1. For the first frame after a software reset, the specified line notification signal is informed at the first scanline regardless of the setting of these bits.

The VPOS interrupt request can be generated only in graphic 2.

SEL Bit (CLUT Control)

This bit selects the color look-up table (CLUT) to be used in graphics data interface block. The CLUT memory can be rewritten any time regardless of the setting in this bit, and the rewritten value is not to be synchronized with the VS signal and to be reflected to the internal operations immediately.

When reflecting the contents of the CLUT memory to the internal operations on the assertion of the VS signal, rewrite the data of the CLUT memory which is not selected and change the value of the bit after completion of the rewriting.

51.2.26 Graphic n Status Monitor Register (GRnMON) (n = 1, 2)

Address(es): GLCDC.GR1MON 000E 1154h, GLCDC.GR2MON 000E 1254h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	UFST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARCST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ARCST	Alpha Blending Status Flag	0: Fade-in/fade-out not in progress for rectangular alpha blending 1: Fade-in/fade-out in progress for rectangular alpha blending	R
b15 to b1	—	Reserved	These bits are read as 0.	R
b16	UFST	Underflow Status Flag	0: No underflow has occurred 1: Underflow has occurred	R
b31 to b17	—	Reserved	These bits are read as 0.	R

ARCST Flag (Alpha Blending Status Flag)

This flag indicates whether or not fade-in or fade-out is in progress for rectangular alpha blending. When the GRnAB1.ARCON bit is 1 (rectangular alpha blending) and the BGEN.VEN bit or GRnVEN.VEN bit is set to 1 (enable reflection of register values on assertion of the VS signal), then this flag becomes 1 at the immediate assertion of the VS signal. When the GRnAB1.ARCON bit is 0 (per-pixel alpha blending) or when the fade-in or fade-out is completed (alpha value reaches the FFh or 00h), this flag becomes 0. If the value of the GRnAB6.ARCCOEF[8:0] bits are 000h and the value of the GRnAB7.ARCDEF[7:0] bits are any value other than FFh or 00h, the alpha value does not reach the FFh or 00h, thus this flag remains 1 (timeout processing is not performed).

UFST Flag (Underflow Status Flag)

This flag becomes 1 when data supply required in the graphics data interface block was delayed (occurrence of an underflow).

The STMON.GRnUF flag becomes 1 when an underflow occurs, and retains its value until it is cleared by software. However, this flag automatically becomes 0 when the graphics data interface block is initialized on assertion of the VS signal.

Even when the current graphics data is not required (the GRnAB1.DISPSEL[1:0] bits are 00b (background color) or 01b (lower-layer graphics)), this flag becomes 1 during the period from the start of the graphics area to the assertion timing of the VS signal for next frame. In addition, for the first frame after a software reset, this flag may become 1 before the VS signal is asserted depending on the value set in the BGSYNC.VP[3:0] bits.

51.2.27 Gamma Correction x Block Register Update Control Register (GAMxVEN) (x = G, B, R)

Address(es): GLCDC.GAMGVEN 000E 1300h, GLCDC.GAMBVEN 000E 1340h, GLCDC.GAMRVEN 000E 1380h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VEN	Gamma Correction x Block Register Value Reflection Enable	0: Disable the reflection of the register values 1: Enable the reflection of the register values (This bit automatically becomes 0)	R/W
b31 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

VEN Bit (Gamma Correction x Block Register Value Reflection Enable)

This bit enables or disables reflection of the register values to the gamma correction x block internal operations on assertion of the VS (vertical synchronization) signal.

When this bit is set to 1, the values of registers GAMSW, GAMxLUT1 to GAMxLUT8, and GAMxAREA1 to GAMxAREA5 are reflected to the internal operations on assertion of the immediate VS signal, and then this bit automatically becomes 0. When the BGEN.VEN bit is 1, the register values are reflected to the internal operations on assertion of the VS signal, regardless of the value of this bit.

Set this bit to 1 only when the BGEN.VEN bit and this bit are both 0. Operation is not guaranteed when setting this bit to 1 while one of the bits is 1.

While the BGEN.VEN bit or this bit is 1, do not modify registers GAMSW, GAMxLUT1 to GAMxLUT8, and GAMxAREA1 to GAMxAREA5. Otherwise, operation is not guaranteed.

51.2.28 Gamma Correction Block Function Switch Register (GAMSW)

Address(es): GLCDC.GAMSW 000E 1304h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GAMON
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	GAMON	Gamma Correction Enable	0: Disables gamma correction 1: Enables gamma correction	R/W
b31 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMGVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.29 Gamma Correction x Table Setting Register 1 (GAMxLUT1) (x = G, B, R)

Address(es): GLCDC.GAMGLUT1 000E 1308h, GLCDC.GAMBLUT1 000E 1348h, GLCDC.GAMRLUT1 000E 1388h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN01[10:0]	Area 1 Gain Setting	Set the gain value for area 1 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN00[10:0]	Area 0 Gain Setting	Set the gain value for area 0 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

Gamma correction for each color is performed as follows ($n = 0$ to 15):

$$D_{out} = (D_{in} - TH_n) \times GAIN_n + OFFSET_n$$

$$OFFSET_n = OFFSET_{n-1} + (TH_n - TH_{n-1}) \times GAIN_{n-1}$$

D_{in}: Input signal. 10 bits. When the CLKPHASE.FRONTGAM bit is 1 (Process gamma correction prior to brightness/contrast correction), 00b is added to the lower 2 bits of the output data of graphic 2 for extension (refer to Figure 51.8).

D_{out}: Output signal. 10 bits.

TH_n: Threshold for area n. 10 bits (set in the GAMxAREAi registers ($i = 1$ to 5)). Up to 15 can be set ($TH_0 = 0$).

GAIN_n: Gain for area n. 0/1024 to 2047/1024 for each area (set in the GAMxLUTj registers ($j = 1$ to 8)). Can be set for up to 16 areas.

OFFSET_n: Offset value for area n. 21 bits (result of internal calculation; up to 15 points ($OFFSET_0 = 0$)). The values are automatically calculated from the assertion of the VS signal to the start of active pixels.

Note: Because the gain is a positive number (≥ 0), the correction curve shows a monotonic increase.

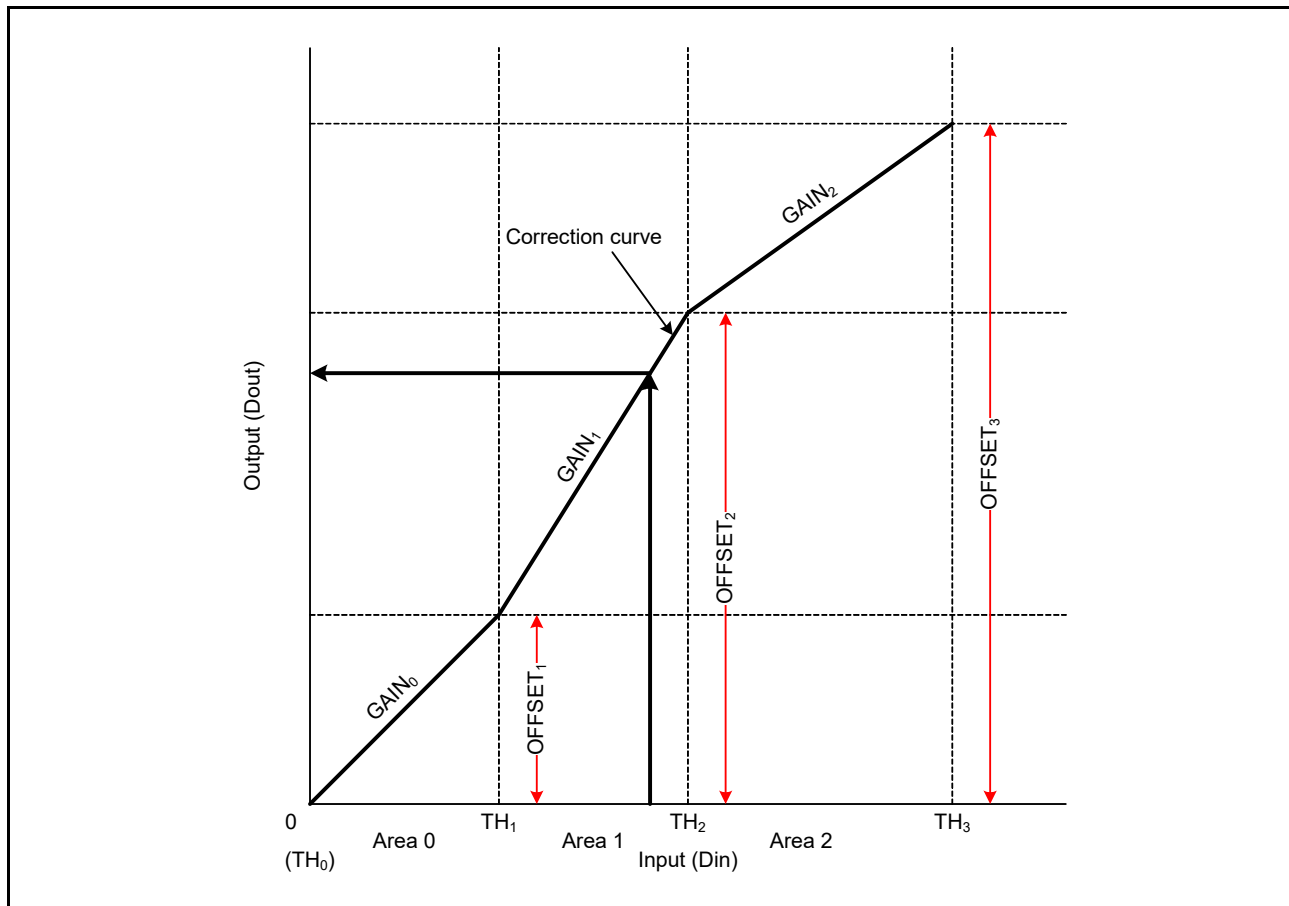


Figure 51.11 Calculation of Gamma Correction Value

51.2.30 Gamma Correction x Table Setting Register 2 (GAMxLUT2) (x = G, B, R)

Address(es): GLCDC.GAMGLUT2 000E 130Ch, GLCDC.GAMBLUT2 000E 134Ch, GLCDC.GAMRLUT2 000E 138Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN03[10:0]	Area 3 Gain Setting	Set the gain value for area 3 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN02[10:0]	Area 2 Gain Setting	Set the gain value for area 2 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.31 Gamma Correction x Table Setting Register 3 (GAMxLUT3) (x = G, B, R)

Address(es): GLCDC.GAMGLUT3 000E 1310h, GLCDC.GAMBLUT3 000E 1350h, GLCDC.GAMRLUT3 000E 1390h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN05[10:0]	Area 5 Gain Setting	Set the gain value for area 5 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN04[10:0]	Area 4 Gain Setting	Set the gain value for area 4 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.32 Gamma Correction x Table Setting Register 4 (GAMxLUT4) (x = G, B, R)

Address(es): GLCDC.GAMGLUT4 000E 1314h, GLCDC.GAMBLUT4 000E 1354h, GLCDC.GAMRLUT4 000E 1394h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN07[10:0]	Area 7 Gain Setting	Set the gain value for area 7 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN06[10:0]	Area 6 Gain Setting	Set the gain value for area 6 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.33 Gamma Correction x Table Setting Register 5 (GAMxLUT5) (x = G, B, R)

Address(es): GLCDC.GAMGLUT5 000E 1318h, GLCDC.GAMBLUT5 000E 1358h, GLCDC.GAMRLUT5 000E 1398h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN09[10:0]	Area 9 Gain Setting	Set the gain value for area 9 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN08[10:0]	Area 8 Gain Setting	Set the gain value for area 8 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.34 Gamma Correction x Table Setting Register 6 (GAMxLUT6) (x = G, B, R)

Address(es): GLCDC.GAMGLUT6 000E 131Ch, GLCDC.GAMBLUT6 000E 135Ch, GLCDC.GAMRLUT6 000E 139Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN11[10:0]	Area 11 Gain Setting	Set the gain value for area 11 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN10[10:0]	Area 10 Gain Setting	Set the gain value for area 10 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.35 Gamma Correction x Table Setting Register 7 (GAMxLUT7) (x = G, B, R)

Address(es): GLCDC.GAMGLUT7 000E 1320h, GLCDC.GAMBLUT7 000E 1360h, GLCDC.GAMRLUT7 000E 13A0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN13[10:0]	Area 13 Gain Setting	Set the gain value for area 13 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN12[10:0]	Area 12 Gain Setting	Set the gain value for area 12 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.36 Gamma Correction x Table Setting Register 8 (GAMxLUT8) (x = G, B, R)

Address(es): GLCDC.GAMGLUT8 000E 1324h, GLCDC.GAMBLUT8 000E 1364h, GLCDC.GAMRLUT8 000E 13A4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

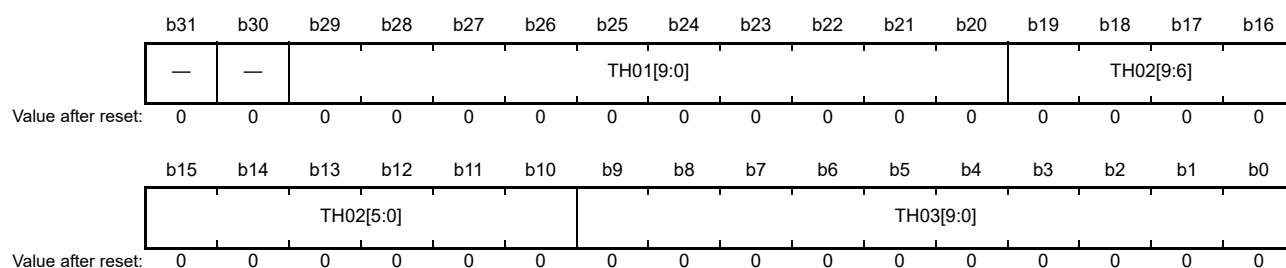
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	GAIN15[10:0]	Area 15 Gain Setting	Set the gain value for area 15 in 11-bit unsigned fixed point number. The location of the decimal point is between b10 and b9. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	GAIN14[10:0]	Area 14 Gain Setting	Set the gain value for area 14 in 11-bit unsigned fixed point number. The location of the decimal point is between b26 and b25. 000h: 0.000 (0/1024) : 400h: 1.000 (1024/1024) : 7FFh: 1.999 (2047/1024)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.37 Gamma Correction x Area Setting Register 1 (GAMxAREA1) (x = G, B, R)

Address(es): GLCDC.GAMGAREA1 000E 1328h, GLCDC.GAMBAREA1 000E 1368h, GLCDC.GAMRAREA1 000E 13A8h



Bit	Symbol	Bit Name	Description	R/W
b9 to b0	TH03[9:0]	Area 3 Start Threshold Setting	Set the start threshold (intensity for each color) of area 3 in 10-bit unsigned integer.	R/W
b19 to b10	TH02[9:0]	Area 2 Start Threshold Setting	Set the start threshold (intensity for each color) of area 2 in 10-bit unsigned integer.	R/W
b29 to b20	TH01[9:0]	Area 1 Start Threshold Setting	Set the start threshold (intensity for each color) of area 1 in 10-bit unsigned integer.	R/W
b31, b30	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

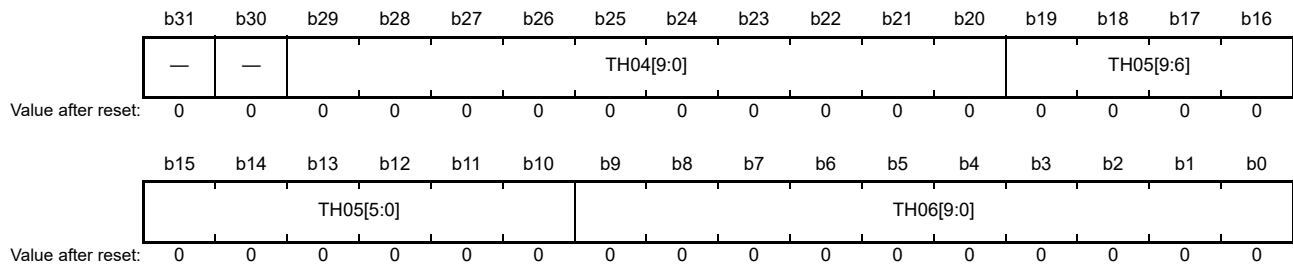
Set the start threshold TH_n of area n ($n = 0$ to 15) to satisfy the condition $TH_n < TH_{n+1}$ ($TH_0 = 000h$, $TH_{16} = 3FFh$). Otherwise, operation is not guaranteed.

Only if TH_n is set to $3FFh$, the setting of $TH_n = TH_{n+1}$ is available.

For details on calculation of the gamma correction value, refer to section 51.2.29, Gamma Correction x Table Setting Register 1 (GAMxLUT1) (x = G, B, R), and Figure 51.11.

51.2.38 Gamma Correction x Area Setting Register 2 (GAMxAREA2) (x = G, B, R)

Address(es): GLCDC.GAMGAREA2 000E 132Ch, GLCDC.GAMBAREA2 000E 136Ch, GLCDC.GAMRAREA2 000E 13ACh

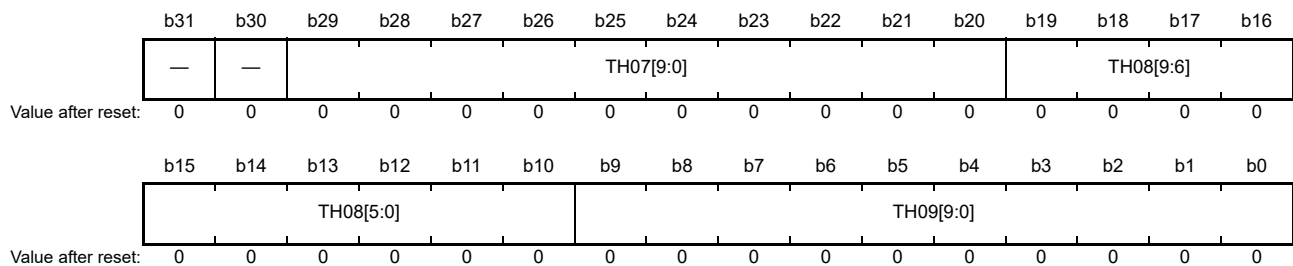


Bit	Symbol	Bit Name	Description	R/W
b9 to b0	TH06[9:0]	Area 6 Start Threshold Setting	Set the start threshold (intensity for each color) of area 6 in 10-bit unsigned integer.	R/W
b19 to b10	TH05[9:0]	Area 5 Start Threshold Setting	Set the start threshold (intensity for each color) of area 5 in 10-bit unsigned integer.	R/W
b29 to b20	TH04[9:0]	Area 4 Start Threshold Setting	Set the start threshold (intensity for each color) of area 4 in 10-bit unsigned integer.	R/W
b31, b30	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.39 Gamma Correction x Area Setting Register 3 (GAMxAREA3) (x = G, B, R)

Address(es): GLCDC.GAMGAREA3 000E 1330h, GLCDC.GAMBAREA3 000E 1370h, GLCDC.GAMRAREA3 000E 13B0h

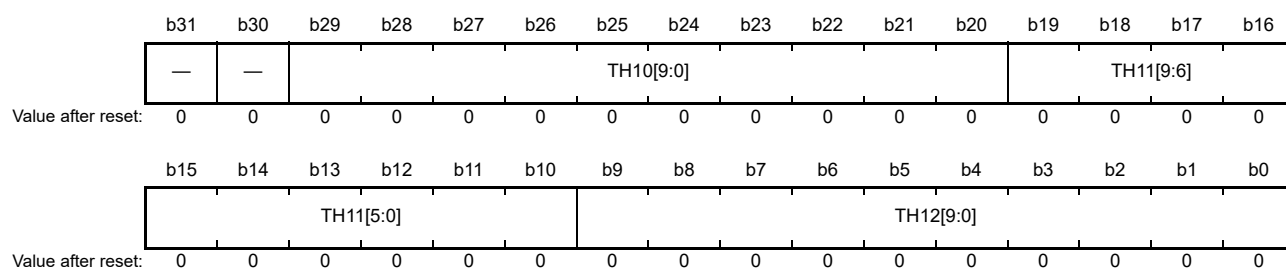


Bit	Symbol	Bit Name	Description	R/W
b9 to b0	TH09[9:0]	Area 9 Start Threshold Setting	Set the start threshold (intensity for each color) of area 9 in 10-bit unsigned integer.	R/W
b19 to b10	TH08[9:0]	Area 8 Start Threshold Setting	Set the start threshold (intensity for each color) of area 8 in 10-bit unsigned integer.	R/W
b29 to b20	TH07[9:0]	Area 7 Start Threshold Setting	Set the start threshold (intensity for each color) of area 7 in 10-bit unsigned integer.	R/W
b31, b30	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.40 Gamma Correction x Area Setting Register 4 (GAMxAREA4) (x = G, B, R)

Address(es): GLCDC.GAMGAREA4 000E 1334h, GLCDC.GAMBAREA4 000E 1374h, GLCDC.GAMRAREA4 000E 13B4h

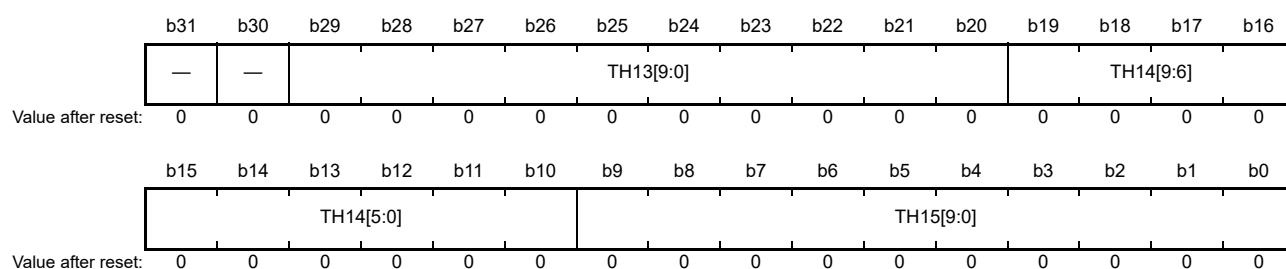


Bit	Symbol	Bit Name	Description	R/W
b9 to b0	TH12[9:0]	Area 12 Start Threshold Setting	Set the start threshold (intensity for each color) of area 12 in 10-bit unsigned integer.	R/W
b19 to b10	TH11[9:0]	Area 11 Start Threshold Setting	Set the start threshold (intensity for each color) of area 11 in 10-bit unsigned integer.	R/W
b29 to b20	TH10[9:0]	Area 10 Start Threshold Setting	Set the start threshold (intensity for each color) of area 10 in 10-bit unsigned integer.	R/W
b31, b30	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.41 Gamma Correction x Area Setting Register 5 (GAMxAREA5) (x = G, B, R)

Address(es): GLCDC.GAMGAREA5 000E 1338h, GLCDC.GAMBAREA5 000E 1378h, GLCDC.GAMRAREA5 000E 13B8h



Bit	Symbol	Bit Name	Description	R/W
b9 to b0	TH15[9:0]	Area 15 Start Threshold Setting	Set the start threshold (intensity for each color) of area 15 in 10-bit unsigned integer.	R/W
b19 to b10	TH14[9:0]	Area 14 Start Threshold Setting	Set the start threshold (intensity for each color) of area 14 in 10-bit unsigned integer.	R/W
b29 to b20	TH13[9:0]	Area 13 Start Threshold Setting	Set the start threshold (intensity for each color) of area 13 in 10-bit unsigned integer.	R/W
b31, b30	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the GAMxVEN.VEN bit or the BGEN.VEN bit is 1.

51.2.42 Output Control Block Register Update Control Register (OUTVEN)

Address(es): GLCDC.OUTVEN 000E 13C0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VEN	Output Control Block Register Value Reflection Enable	0: Disable the reflection of the register values 1: Enable the reflection of the register values (This bit automatically becomes 0)	R/W
b31 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

VEN Bit (Output Control Block Register Value Reflection Enable)

This bit enables or disables reflection of the register values to the output control block internal operations on assertion of the VS (vertical synchronization) signal.

When this bit is set to 1, the values of registers BRIGHT1, BRIGHT2, CONTRAST, PANELDTHA, and CLKPHASE are reflected to the internal operations on assertion of the immediate VS signal, and then this bit automatically becomes 0. When the BGEN.VEN bit is 1, the register values are reflected to the internal operations on assertion of the VS signal, regardless of the value of this bit.

Set this bit to 1 only when the BGEN.VEN bit and this bit are both 0. Operation is not guaranteed when setting this bit to 1 while one of the bits is 1.

While the BGEN.VEN bit or this bit is 1, do not modify registers BRIGHT1, BRIGHT2, CONTRAST, PANELDTHA, and CLKPHASE. Otherwise, operation is not guaranteed.

51.2.43 Output Interface Register (OUTSET)

Address(es): GLCDC.OUTSET 000E 13C4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	ENDIANON	—	—	—	SWAPON	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	FORMAT[1:0]	—	—	FRQSEL	—	—	—	—	—	DIRSEL	—	—	PHASE[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	PHASE[1:0]	Serial RGB Data Output Delay Control	b1 b0 0 0: 0 cycles 0 1: 1 cycle 1 0: 2 cycles 1 1: 3 cycles	R/W
b3, b2	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b4	DIRSEL	Serial RGB Scan Direction Select	0: Forward scan 1: Reverse scan	R/W
b7 to b5	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b8	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b9	FRQSEL	Pixel Clock Division Control*1	0: No division (parallel RGB) 1: Divide-by-4 (serial RGB)	R/W
b11, b10	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b13, b12	FORMAT[1:0]	Output Data Format Select*2	b13 b12 0 0: RGB(888) 0 1: RGB(666) 1 0: RGB(565) 1 1: Serial RGB	R/W
b23 to b14	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b24	SWAPON	Pixel Order Control	0: R-G-B order 1: B-G-R order	R/W
b27 to b25	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b28	ENDIANON	Bit Endian Control	0: Little endian 1: Big endian	R/W
b31 to b29	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

Note 1. Set the same value as the PANELCLK.PIXSEL bit.

Note 2. Match the contents of the settings of these bits with the those of the PANELDTHA.FORM[1:0] bits. However, if these bits are 11b (serial RGB), set the FORM[1:0] bits to 00b.

This register specifies the format of LCD_DATA23 to LCD_DATA0 to be output from the data format conversion block. For the relation between the settings for each bit except the FRQSEL bit and assignment of the output pins (LCD_DATA23 to LCD_DATA0), refer to Table 51.4 to Table 51.8. In addition, refer to Figure 51.7 for the value set in the PHASE[1:0] bits and output timing of the data.

PHASE[1:0] Bits (Serial RGB Data Output Delay Control)

These bits control data output delay in serial RGB format. When these bits are 00b (0 cycles), pixel data (R, B, or invalid data, depending on the setting in this register) is output one pixel clock (PXCLK) after the HE (horizontal data enable) signal. When these bits are any value other than 00b, pixel data output is delayed for a specified number of LCD_CLK

cycles.

DIRSEL Bit (Serial RGB Scan Direction Select)

This bit controls the assignment of invalid data for the serial RGB format. When this bit is 1 (reverse scan), invalid data is output before valid data. When the bit is 0 (forward scan), invalid data is output after valid data.

FRQSEL Bit (Pixel Clock Division Control)

This bit specifies division ratio of the PXCLK (pixel clock) for the LCD_CLK (panel clock). Set this bit to 1 to set the PXCLK frequency to a quarter of the LCD_CLK frequency only when the OUTSET.FORMAT[1:0] bits are 11b (serial RGB format). Set this bit to 0 to set the PXCLK frequency to the same as the LCD_CLK frequency when the OUTSET.FORMAT[1:0] bits are 00b, 01b, or 10b (parallel RGB format). Otherwise, operation is not guaranteed.

FORMAT[1:0] Bits (Output Data Format Select)

These bits select the format of RGB data output to LCD panel. Set the value of the PANELDTHA.FORM[1:0] bits in accordance with the format selected in these bits. When these bits are set to 11b (serial RGB format), set the PANELDTHA.FORM[1:0] bits to 00b (RGB(888)). Otherwise, operation is not guaranteed.

SWAPON Bit (Pixel Order Control)

This bit controls the pixel order of RGB data output to LCD panel. When this bit is set to 1, data is assigned to the output pins in B-G-R order, and when this bit is 0, data is assigned in the R-G-B order. In the parallel RGB format, data is assigned to the output pins in a preset order from the MSB side (LCD_DATA23). In the serial RGB format, data is assigned in order of time.

ENDIANON Bit (Bit Endian Control)

This bit controls the bit order of RGB data output to LCD panel. When this bit is 1, data is assigned to the output pins in big endian (LCD_DATA23 side to the least significant bit of R, G, or B), and when this bit is 0, data is assigned in little endian (LCD_DATA23 side to the most significant bit of R, G, or B).

51.2.44 Brightness Adjustment Register 1 (BRIGHT1)

Address(es): GLCDC.BRIGHT1 000E 13C8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b9 to b0	BRTG[9:0]	G Channel Brightness Adjustment Value Setting	Set the brightness adjustment value for G channel with +512 offset. 000h: –512 : 200h: 0 : 3FFh: +511	R/W
b31 to b10	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the OUTVEN.VEN bit or the BGEN.VEN bit is 1.

Brightness adjustment for the G channel is performed as follows:

$$G_{out} = G_{in} + BRTG[9:0] - 512$$

Gout: Output of brightness adjustment (input of contrast adjustment); 10-bit unsigned integer

Gin: Input of brightness adjustment; 10-bit unsigned integer

51.2.45 Brightness Adjustment Register 2 (BRIGHT2)

Address(es): GLCDC.BRIGHT2 000E 13CCh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b9 to b0	BRTR[9:0]	R Channel Brightness Adjustment Value Setting	Set the brightness adjustment value for R channel with +512 offset. 000h: –512 : 200h: 0 : 3FFh: +511	R/W
b15 to b10	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b25 to b16	BRTB[9:0]	B Channel Brightness Adjustment Value Setting	Set the brightness adjustment value for B channel with +512 offset. 000h: –512 : 200h: 0 : 3FFh: +511	R/W
b31 to b26	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the OUTVEN.VEN bit or the BGEN.VEN bit is 1.

Brightness adjustment for the R channel is performed as follows:

$$R_{out} = R_{in} + BRTR[9:0] - 512$$

R_{out}: Output of brightness adjustment (input of contrast adjustment); 10-bit unsigned integer

R_{in}: Input of brightness adjustment; 10-bit unsigned integer

Brightness adjustment for the B channel is performed as follows:

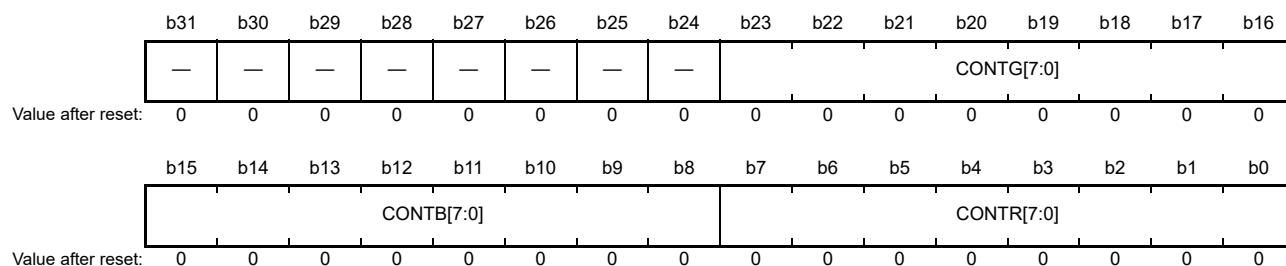
$$B_{out} = B_{in} + BRTB[9:0] - 512$$

B_{out}: Output of brightness adjustment (input of contrast adjustment); 10-bit unsigned integer

B_{in}: Input of brightness adjustment; 10-bit unsigned integer

51.2.46 Contrast Adjustment Register (CONTRAST)

Address(es): GLCDC.CONTRAST 000E 13D0h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	CONTR[7:0]	R Channel Contrast Adjustment Value Setting	Set the contrast adjustment value for R channel in 8-bit unsigned fixed point number. The location of the decimal point is between b7 and b6. 00h: 0/128 = 0.000 : 80h: 128/128 = 1.000 : FFh: 255/128 = 1.992	R/W
b15 to b8	CONTB[7:0]	B Channel Contrast Adjustment Value Setting	Set the contrast adjustment value for B channel in 8-bit unsigned fixed point number. The location of the decimal point is between b15 and b14. 00h: 0/128 = 0.000 : 80h: 128/128 = 1.000 : FFh: 255/128 = 1.992	R/W
b23 to b16	CONTG[7:0]	G Channel Contrast Adjustment Value Setting	Set the contrast adjustment value for G channel in 8-bit unsigned fixed point number. The location of the decimal point is between b23 and b22. 00h: 0/128 = 0.000 : 80h: 128/128 = 1.000 : FFh: 255/128 = 1.992	R/W
b31 to b24	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the OUTVEN.VEN bit or the BGEN.VEN bit is 1.

Contrast adjustment of each channel is performed as follows (x = R, G, B):

$$Dx_{out} = Dx_{in} \times CONTx[7:0]$$

Dx_{out}: Output of contrast adjustment; 10-bit unsigned integer

Dx_{in}: Input of contrast adjustment (output of brightness adjustment); 10-bit unsigned integer

51.2.47 Panel Dither Control Register (PANELDTHA)

Address(es): GLCDC.PANELDTHA 000E 13D4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	SEL[1:0]	—	—	—	FORM[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	PA[1:0]	—	—	—	PB[1:0]	—	—	—	PC[1:0]	—	—	—	PD[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	PD[1:0]	Dither Pattern Value D Setting	Set the pattern value D of 2×2 pattern dither. The recommended value is 01b.	R/W
b3, b2	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b5, b4	PC[1:0]	Dither Pattern Value C Setting	Set the pattern value C of 2×2 pattern dither. The recommended value is 10b.	R/W
b7, b6	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b9, b8	PB[1:0]	Dither Pattern Value B Setting	Set the pattern value B of 2×2 pattern dither. The recommended value is 00b.	R/W
b11, b10	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b13, b12	PA[1:0]	Dither Pattern Value A Setting	Set the pattern value A of 2×2 pattern dither. The recommended value is 11b.	R/W
b15, b14	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b17, b16	FORM[1:0]	Output Format Select*1	b17 b16 0 0: RGB(888) 0 1: RGB(666) 1 0: RGB(565) 1 1: Setting prohibited	R/W
b19, b18	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b21, b20	SEL[1:0]	Rounding Mode Setting	b21 b20 0 0: Truncate 0 1: Round-off 1 0: 2×2 pattern dither 1 1: Setting prohibited	R/W
b31 to b22	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the OUTVEN.VEN bit or the BGEN.VEN bit is 1.

Note 1. Match the contents of the settings of these bits with the those of the OUTSET.FORMAT[1:0] bits. However, if the FORMAT[1:0] bits are 11b (serial RGB), set these bits to 00b.

This register controls the operation of the dithering block. Figure 51.12 shows the structure of the dithering block.

FORM[1:0] Bits (Output Format Select)

These bits specify the position for the execution of rounding in the dithering block. These bits must be set in accordance with the OUTSET.FORMAT[1:0] bits. When the OUTSET.FORMAT[1:0] bits are 11b (serial RGB), set these bits to 00b (RGB(888)). Otherwise, operation is not guaranteed.

SEL[1:0] Bits (Rounding Mode Setting)

These bits specify the dither operation mode. Dithering is performed for the bits whose color depth is below the level specified in the FORM[1:0] bits. The dither pattern set in the PANELDTHA.PA[1:0], PB[1:0], PC[1:0], and PD[1:0] bits are used for 2×2 pattern dither.

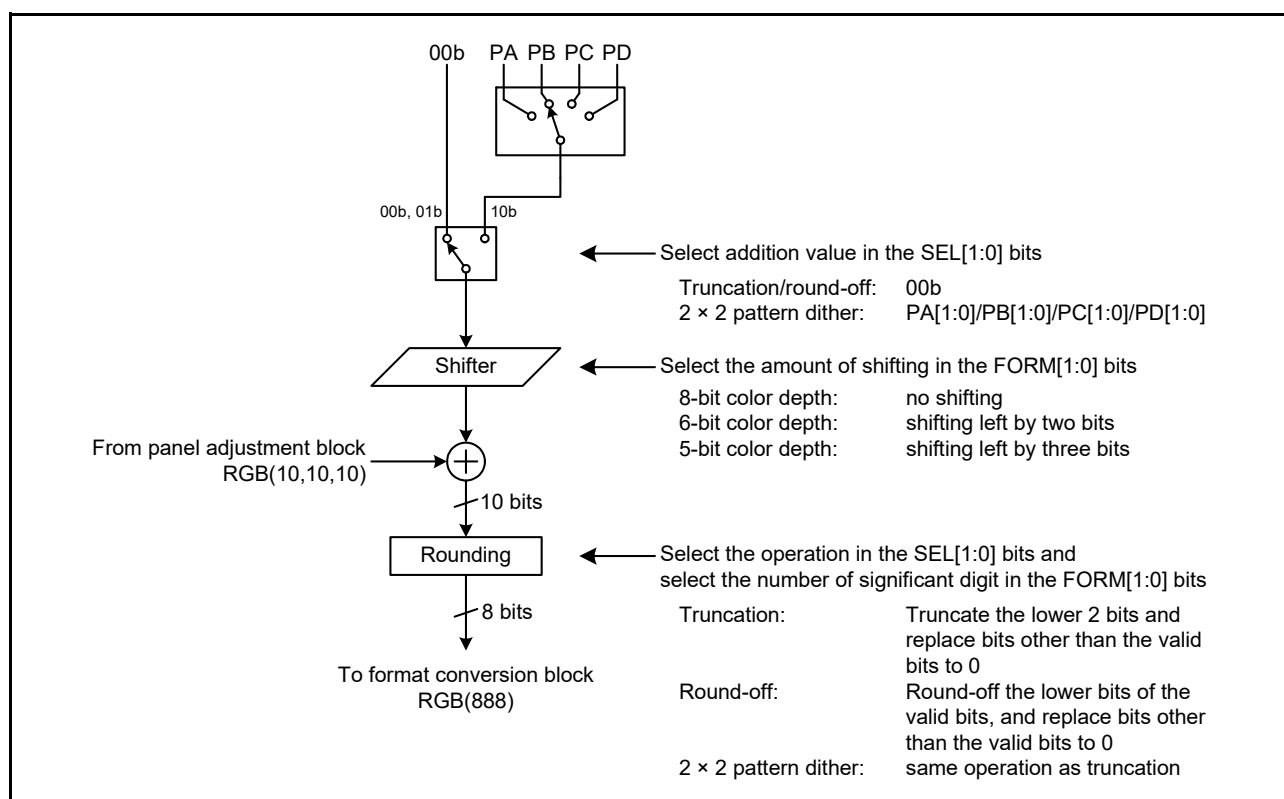


Figure 51.12 Structure of Dithering Block

Figure 51.13 shows the addition value selection method for 2×2 pattern dither.

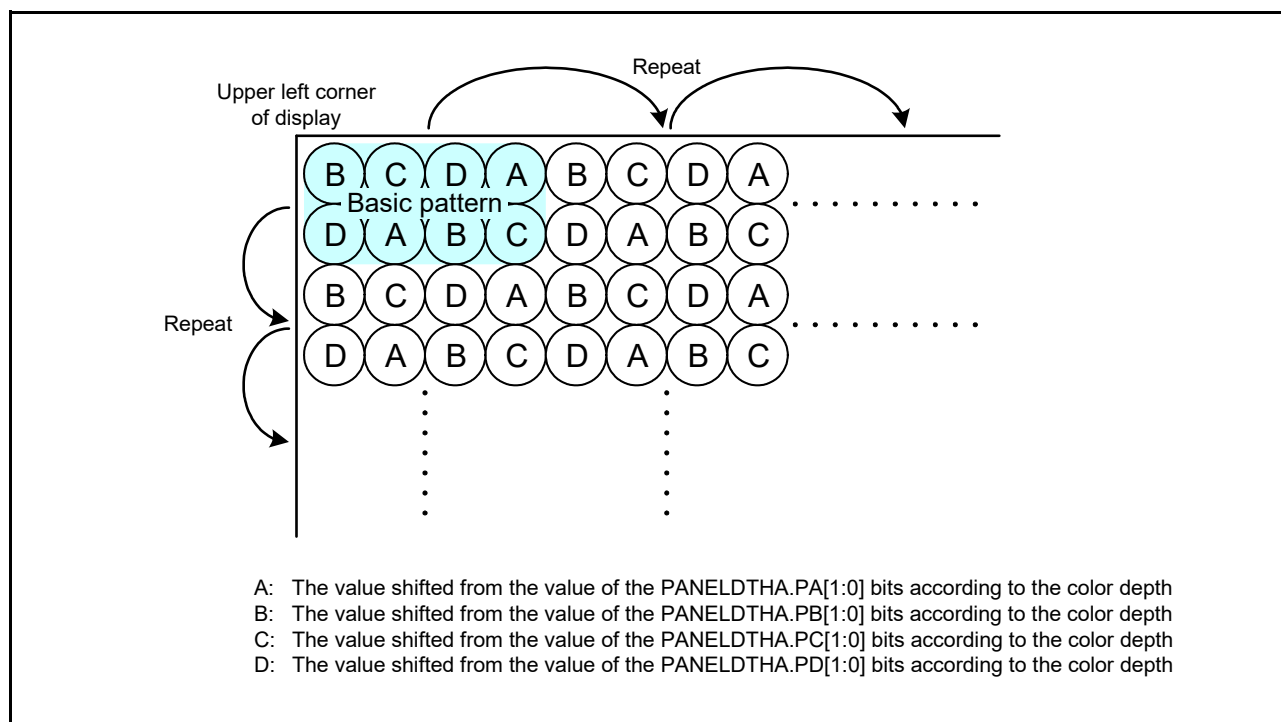
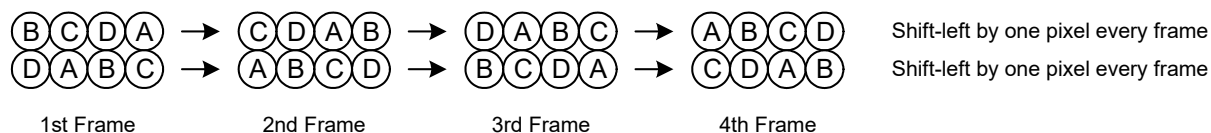


Figure 51.13 Addition Value Selection Method for 2×2 Pattern Dither

The basic pattern is shifted within a frame as shown below and this is repeated every four frames.



When 2×2 pattern dither is used, the active display area of the background screen must be an integer multiple of the basic pattern (a multiple of 4 for horizontal and a multiple of 2 for vertical). If serial RGB is selected as the output format, set [horizontal active pixels (multiple of 4) + 2] to the BGHSIZE.HW[10:0] bits.

51.2.48 Output Phase Control Register (CLKPHASE)

Address(es): GLCDC.CLKPHASE 000E 13E4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	FRONT GAM	—	—	—	LCDE D G	—	TCON0 EDG	TCON1 EDG	TCON2 EDG	TCON3 EDG	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b3	TCON3EDG	TCON3 Output Phase Control	0: The output of LCD_TCON3 is synchronized with rising edges of LCD_CLK 1: The output of LCD_TCON3 is synchronized with falling edges of LCD_CLK	R/W
b4	TCON2EDG	TCON2 Output Phase Control	0: The output of LCD_TCON2 is synchronized with rising edges of LCD_CLK 1: The output of LCD_TCON2 is synchronized with falling edges of LCD_CLK	R/W
b5	TCON1EDG	TCON1 Output Phase Control	0: The output of LCD_TCON1 is synchronized with rising edges of LCD_CLK 1: The output of LCD_TCON1 is synchronized with falling edges of LCD_CLK	R/W
b6	TCON0EDG	TCON0 Output Phase Control	0: The output of LCD_TCON0 is synchronized with rising edges of LCD_CLK 1: The output of LCD_TCON0 is synchronized with falling edges of LCD_CLK	R/W
b7	—	Reserved	This bit is read as 0. Writing to this bit has no effect.	R
b8	LCDEEDG	DATA Output Phase Control	0: The outputs of LCD_DATA are synchronized with rising edges of LCD_CLK 1: The outputs of LCD_DATA are synchronized with falling edges of LCD_CLK	R/W
b11 to b9	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b12	FRONTGAM	Correction Sequence Control	0: Process brightness/contrast adjustment prior to gamma correction 1: Process gamma correction prior to brightness/contrast adjustment	R/W
b31 to b13	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: This register setting is reflected to the internal operations when the VS signal is asserted while the OUTVEN.VEN bit or the BGEN.VEN bit is 1.

FRONTGAM Bit (Correction Sequence Control)

This bit controls the order of processing in the panel adjustment block. When this bit is set to 1, gamma correction is followed by brightness and contrast adjustment, and when it is set to 0, gamma correction follows brightness and contrast adjustment. In both cases, each RGB data output from the graphic 2 block is extended from 8 bits to 10 bits (with 00b appended to the lower side). The output of the panel adjustment block is rounded to 10 bits and is input to the dithering block. Although the sequence of gamma correction and brightness/contrast adjustment can be reversed using this bit, brightness adjustment always precedes contrast adjustment.

51.2.49 Reference Timing Setting Register (TCONTIM)

Address(es): GLCDC.TCONTIM 000E 1404h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	OFFSET[10:0]	Horizontal Synchronization Signal Reference Timing Setting	000h: 1 pixel : 3FFh: 1024 pixels Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	HALF[10:0]	Vertical Synchronization Signal Transition Timing Setting	000h: 1 pixel (no delay) : 3FFh: 1024 pixels Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

OFFSET[10:0] Bits (Horizontal Synchronization Signal Reference Timing Setting)

These bits specify the reference timing position for generating the horizontal synchronization signal in the TCON block. Set the offset from the internal HS (horizontal synchronization) signal in pixels. Figure 51.14 shows the horizontal synchronization signal generation timing in the TCON block.

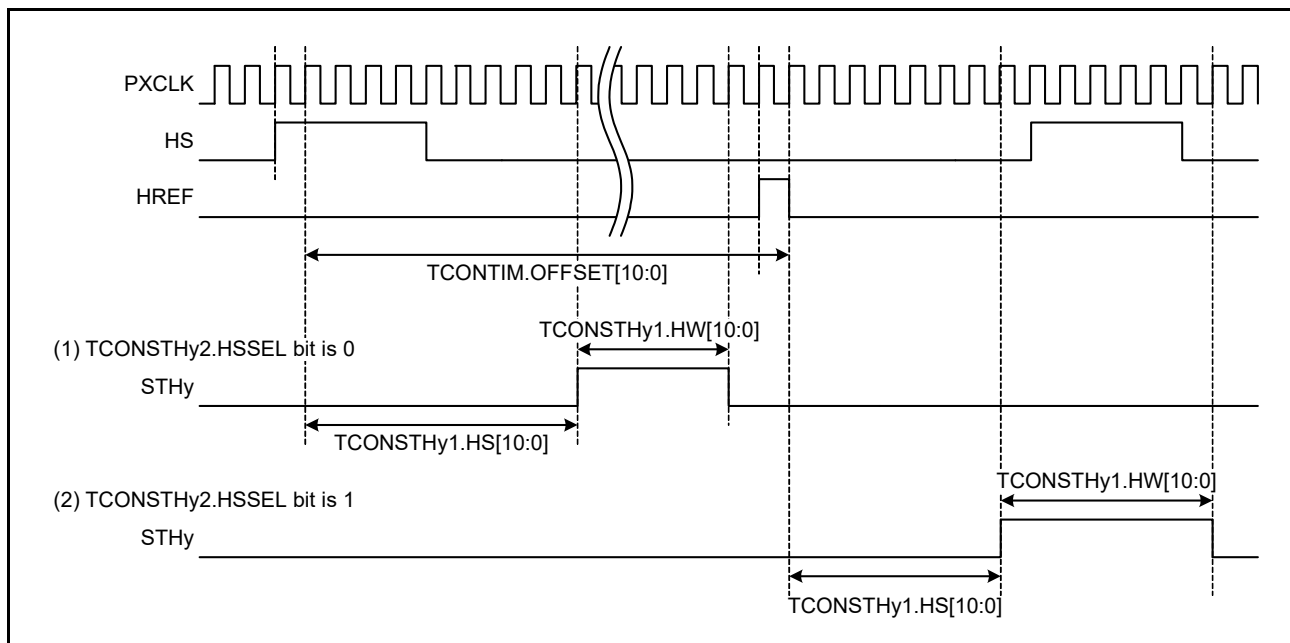


Figure 51.14 Horizontal Synchronization Signal Generation Timing in the TCON Block (y = A, B)

HALF[10:0] Bits (Vertical Synchronization Signal Transition Timing Setting)

These bits specify the waveform transition position within a single horizontal line for generating the vertical synchronization signal in the TCON block. Set the transition timing as a delay from the internal HS (horizontal synchronization) signal in pixel units.

When a value other than 000h is set to these bits, the internal line counter runs at twice the speed. In this case, set a value of twice the desired number of lines to the TCONSTV1 register.

Figure 51.15 shows the vertical synchronization signal transition timing in the TCON block.

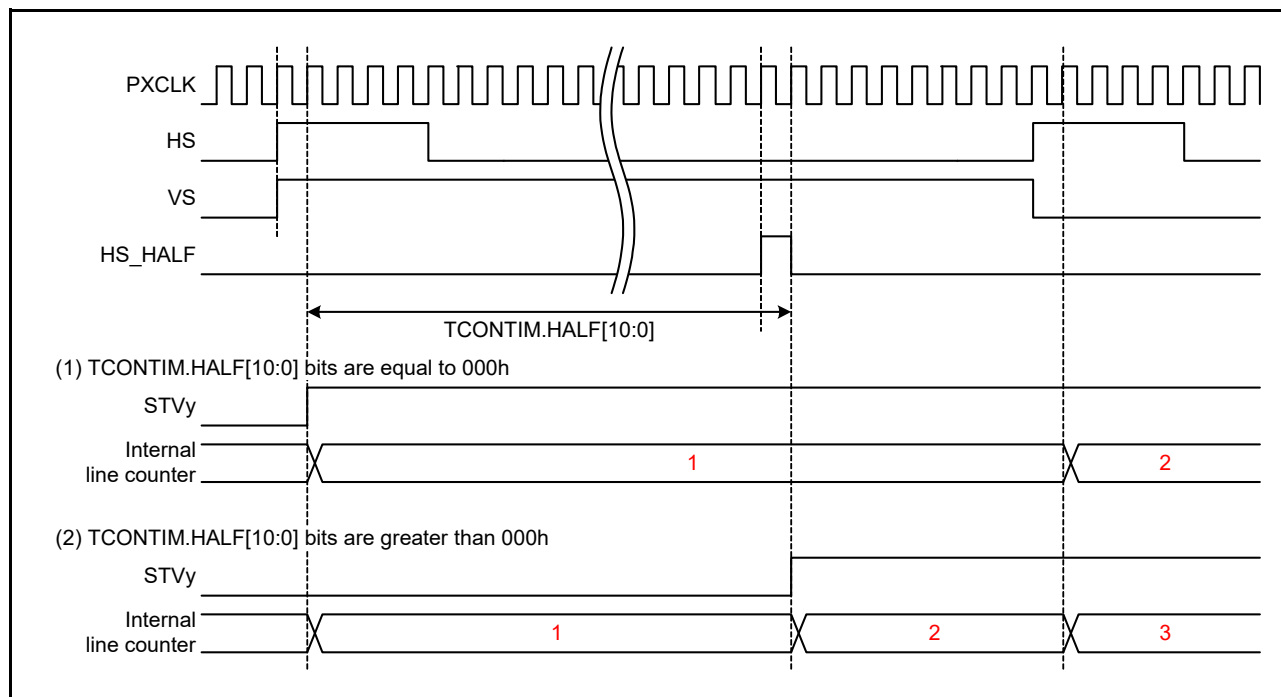


Figure 51.15 Vertical Synchronization Signal Transition Timing in the TCON Block (y = A, B)

Figure 51.16 shows the structure of the output control block and the signal flow.

The pixel data and control signals are output to the external pins along different paths. They are generated based on the same signal and the internal delay is the same 3 PXCLK cycles. Therefore, differences in delay due to the settings of the respective registers become differences in the timing of operations of the external pins.

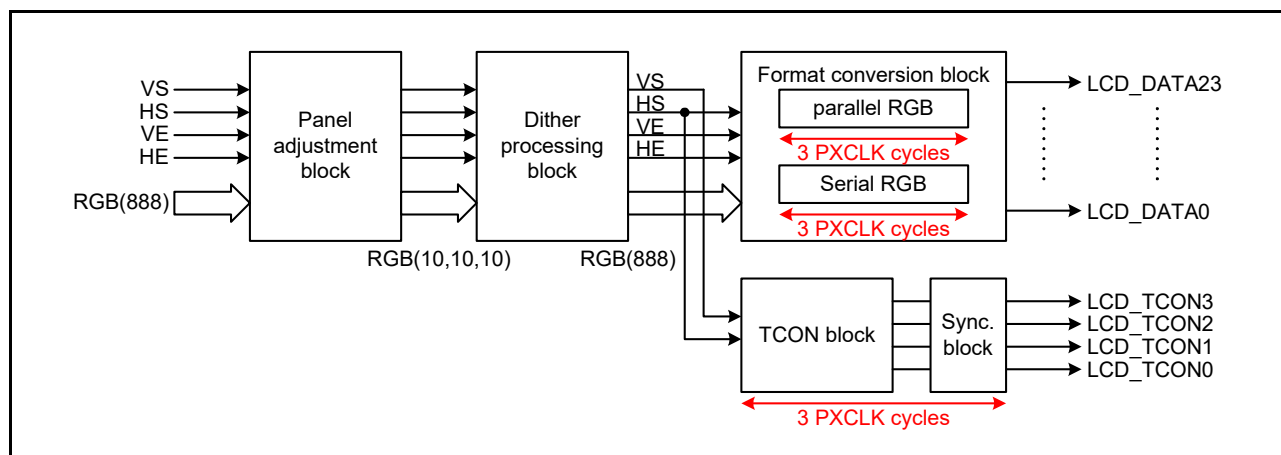


Figure 51.16 Internal Structure of Output Control Block

51.2.50 Vertical Timing Setting Register y1 (TCONSTVy1) (y = A, B)

Address(es): GLCDC.TCONSTVA1 000E 1408h, GLCDC.TCONSTVB1 000E 1410h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	VW[10:0]	STVy Signal Pulse Width Setting*1	000h: 0 lines (0H) (STVy signal is not asserted) : 7FFh: 2047 lines (2047H)	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	VS[10:0]	STVy Signal Assertion Timing Setting*1	000h: 0 lines (0H) (no delay) : 7FFh: 2047 lines (2047H)	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

Note 1. When setting the TCONTIM.HALF[10:0] bits to any value other than 000h, set these bits to twice the desired lines.

VW[10:0] Bits (STVy Signal Pulse Width Setting)

These bits specify the assertion width of the STVy (vertical synchronization) signal generated in the TCON block in line units. When the setting of these bits is 000h, the STVy signal is not asserted.

VS[10:0] Bits (STVy Signal Assertion Timing Setting)

These bits specify the assertion timing of the STVy signal generated in the TCON block as a delay in unit of lines from assertion of the VS signal. The transition position in a scanline is specified in the TCONTIM.HALF[10:0] bits.

Figure 51.17 shows the transition timing of STVy signal to be generated.

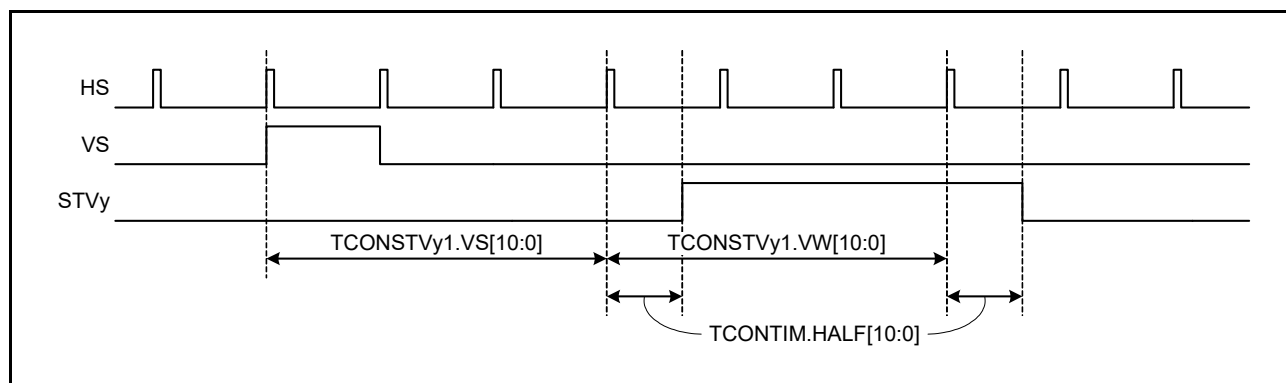


Figure 51.17 Generation of STVy Signal (y = A, B)

51.2.51 Vertical Timing Setting Register y2 (TCONSTV_y2) (y = A, B)

Address(es): GLCDC.TCONSTVA2 000E 140Ch, GLCDC.TCONSTVB2 000E 1414h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	INV	—	SEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SEL[2:0]	Output Signal Select	b2 b0 0 0 0: STVA 0 0 1: STVB 0 1 0: STHA 0 1 1: STHB 1 0 0: Setting prohibited 1 0 1: Setting prohibited 1 1 0: Setting prohibited 1 1 1: DE	R/W
b3	—	Reserved	These bits are read as 0. Writing to this bit has no effect.	R
b4	INV	STV _y Signal Polarity Inversion	0: Do not invert the STV _y signal 1: Invert the STV _y signal	R/W
b31 to b5	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

SEL[2:0] Bits (Output Signal Select)

These bits select the output signal for the LCD_TCON0 and LCD_TCON1 pin. TCONSTVA2.SEL[2:0] bits are used to select signal for LCD_TCON0 pin, and TCONSTVB2.SEL[2:0] bits are for the LCD_TCON1 pin.

Figure 51.18 shows the structure of inversion control and output signal selection.

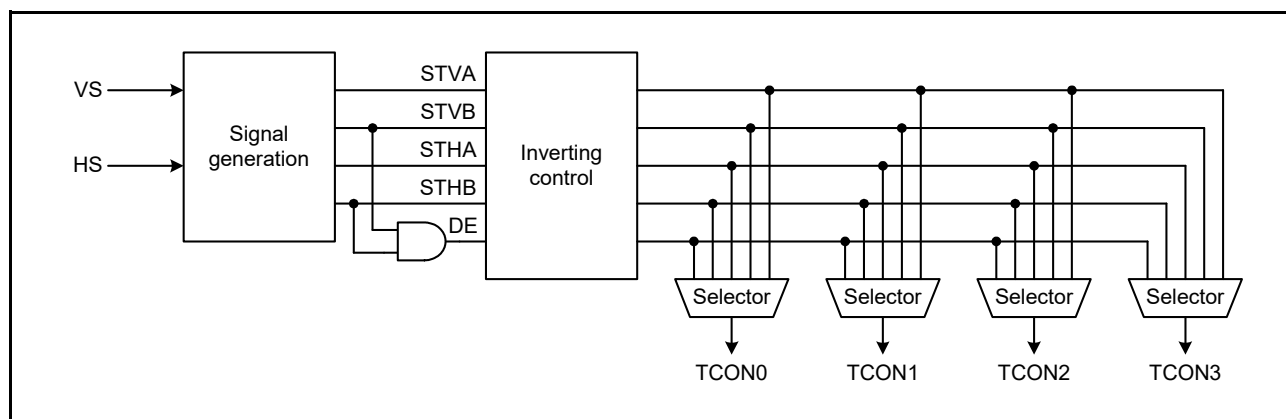


Figure 51.18 Structure of Inversion Control and Output Signal Selects

51.2.52 Horizontal Timing Setting Register y1 (TCONSTHy1) (y = A, B)

Address(es): GLCDC.TCONSTHA1 000E 1418h, GLCDC.TCONSTHB1 000E 1420h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	HW[10:0]	STHy Signal Pulse Width Setting	000h: 0 pixels (STHy signal is not asserted) : 3FFh: 1023 pixels Settings other than the above are prohibited.	R/W
b15 to b11	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b26 to b16	HS[10:0]	STHy Signal Assertion Timing Setting	000h: 0 pixels (no delay) : 3FFh: 1023 pixels Settings other than the above are prohibited.	R/W
b31 to b27	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

HW[10:0] Bits (STHy Signal Pulse Width Setting)

These bits specify the assertion width of the STHy (horizontal synchronization) signal generated in the TCON block in pixel units. When setting these bits to 000h, STHy signal is not asserted.

HS[10:0] Bits (STHy Signal Assertion Timing Setting)

These bits specify the assertion timing of the STHy signal generated in the TCON block as a delay from the HS signal or the offset reference position specified by the TCONTIM.OFFSET[10:0] bits, in unit of pixels.

Figure 51.14 shows the STHy signal generation timing.

Setting the TCONTIM.OFFSET[10:0], HW[10:0], and HS[10:0] bits appropriately leads to the generation of an STHy signal having a width spanning an interval from a point before assertion to a point after negation of the HS signal (spanning the HS signal).

When an STHy signal spanning the HS signal is generated, the horizontal synchronization signal of the last line of the frame extends into the first line of the next frame. In this case, when the BGEN.EN bit is set to 0 to stop operation of the GLCDC, the STHy signal is not negated at the end of the frame and remains asserted up to the point set in the HW[10:0] bits. However, setting the BGEN.SWRST bit to 0 returns the STHy signal back to the initial state immediately.

51.2.53 Horizontal Timing Setting Register y2 (TCONSTHy2) (y = A, B)

Address(es): GLCDC.TCONSTHA2 000E 141Ch, GLCDC.TCONSTHB2 000E 1424h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	HSSEL	—	—	—	INV	—	SEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	SEL[2:0]	Output Signal Select	b2 b0 0 0 0: STVA 0 0 1: STVB 0 1 0: STHA 0 1 1: STHB 1 0 0: Setting prohibited 1 0 1: Setting prohibited 1 1 0: Setting prohibited 1 1 1: DE	R/W
b3	—	Reserved	These bits are read as 0. Writing to this bit has no effect.	R
b4	INV	STHy Signal Polarity Inversion	0: Do not invert the STHy signal 1: Invert the STHy signal	R/W
b7 to b5	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b8	HSSEL	STHy Signal Reference Timing Select	0: Select HS signal as reference for signal generation 1: Select the offset position specified in the TCONTIM.OFFSET[10:0] bits as reference for signal generation	R/W
b31 to b9	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

SEL[2:0] Bits (Output Signal Select)

These bits select the output signal for the LCD_TCON2 and LCD_TCON3 pin. TCONSTHA2.SEL[2:0] bits are used to select signal for LCD_TCON2 pin, and TCONSTHB2.SEL[2:0] bits are for the LCD_TCON3 pin.

HSSEL Bit (STHy Signal Reference Timing Select)

This bit selects the reference timing for generating the STHy (horizontal synchronization) signal. For details on the relation between this bit setting and the generated STHy signal, refer to Figure 51.14.

For the structure of the inversion control and output signal select, refer to Figure 51.18.

51.2.54 Data Enable Polarity Setting Register (TCONDE)

Address(es): GLCDC.TCONDE 000E 1428h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	INV
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	INV	DE Signal Polarity Inversion	0: Do not invert the DE signal 1: Invert the DE signal	R/W
b31 to b1	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

INV Bit (DE Signal Polarity Inversion)

This bit controls polarity inversion of the data enable signal (DE). The DE signal is the logical AND of the STVB and STHB signals.

51.2.55 Status Detection Control Register (DTCTEN)

Address(es): GLCDC.DTCTEN 000E 1440h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	GR2UF DTC	GR1UF DTC	VPOSD TC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VPOSDTC	Specified Line Notification Detection Enable	0: Disable detection of specified line notification 1: Enable detection of specified line notification	R/W
b1	GR1UFDTC	Graphic 1 Underflow Detection Enable	0: Disable detection of graphic 1 underflow 1: Enable detection of graphic 1 underflow	R/W
b2	GR2UFDTC	Graphic 2 Underflow Detection Enable	0: Disable detection of graphic 2 underflow 1: Enable detection of graphic 2 underflow	R/W
b31 to b3	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

VPOSDTC Bit (Specified Line Notification Detection Enable)

This bit enables or disables detection of the specified line notification from graphic 2. When this bit is 1, the STMON.VPOS flag becomes 1 when a specified line notification from graphic 2 is detected. When it is 0, the VPOS flag does not become 1 even when a specified line notification from graphic 2 is detected.

GR1UFDTC Bit (Graphic 1 Underflow Detection Enable)

This bit enables or disables detection of the graphic 1 underflow. When this bit is 1, the STMON.GR1UF flag becomes 1 when an underflow occurs in graphic 1. When it is 0, the GR1UF flag does not become 1 even when an underflow occurs in graphic 1. The underflow status in graphic 1 is automatically cleared on assertion of the VS signal regardless of the value of this bit, and normal operation is recovered.

GR2UFDTC Bit (Graphic 2 Underflow Detection Enable)

This bit enables or disables detection of the graphic 2 underflow. When this bit is 1, the STMON.GR2UF flag becomes 1 when an underflow occurs in graphic 2. When it is 0, the GR2UF flag does not become 1 even when an underflow occurs in graphic 2. The underflow status in graphic 2 is automatically cleared on assertion of the VS signal regardless of the value of this bit, and normal operation is recovered.

51.2.56 Interrupt Request Enable Control Register (INTEN)

Address(es): GLCDC.INTEN 000E 1444h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	GR2UF INTEN	GR1UF INTEN	VPOS INTEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VPOSINTEN	VPOS Interrupt Enable	0: Disable VPOS interrupt request 1: Enable VPOS interrupt request	R/W
b1	GR1UFINTEN	GR1UF Interrupt Enable	0: Disable GR1UF interrupt request 1: Enable GR1UF interrupt request	R/W
b2	GR2UFINTEN	GR2UF Interrupt Enable	0: Disable GR2UF interrupt request 1: Enable GR2UF interrupt request	R/W
b31 to b3	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

VPOSINTEN Bit (VPOS Interrupt Enable)

This bit enables or disables the VPOS interrupt request. When this bit is 1, the VPOS interrupt request is output when the STMON.VPOS flag becomes 1. When it is 0, the VPOS interrupt request is not output even when the VPOS flag becomes 1. If this bit is set to 0 during the VPOS interrupt request is output, the VPOS flag does not change, but the VPOS interrupt request is negated.

GR1UFINTEN Bit (GR1UF Interrupt Enable)

This bit enables or disables the GR1UF interrupt request. When this bit is 1, the GR1UF interrupt request is output when the STMON.GR1UF flag becomes 1. When it is 0, the GR1UF interrupt request is not output even when the GR1UF flag becomes 1. If this bit is set to 0 during the GR1UF interrupt request is output, the GR1UF flag does not change, but the GR1UF interrupt request is negated.

GR2UFINTEN Bit (GR2UF Interrupt Enable)

This bit enables or disables the GR2UF interrupt request. When this bit is 1, the GR2UF interrupt request is asserted when the STMON.GR2UF flag becomes 1. When it is 0, the GR2UF interrupt request is not output even when the GR2UF flag becomes 1. If this bit is set to 0 during the GR2UF interrupt request is output, the GR2UF flag does not change, but the GR2UF interrupt request is negated.

51.2.57 Detected Status Clear Register (STCLR)

Address(es): GLCDC.STCLR 000E 1448h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	GR2UF CLR	GR1UF CLR	VPOS CLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VPOSCLR	VPOS Flag Clear	Writing 1 to this bit clears the STMON.VPOS flag. Writing 0 has no effect. The read value is 0.	R/W
b1	GR1UFCLR	GR1UF Flag Clear	Writing 1 to this bit clears the STMON.GR1UF flag. Writing 0 has no effect. The read value is 0.	R/W
b2	GR2UFCLR	GR2UF Flag Clear	Writing 1 to this bit clears the STMON.GR2UF flag. Writing 0 has no effect. The read value is 0.	R/W
b31 to b3	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

VPOSCLR Bit (VPOS Flag Clear)

Writing 1 to this bit clears the STMON.VPOS flag. This bit is only valid for the VPOS flag, and does not directly affect the other status and interrupt request signals. However, a VPOS interrupt request is negated by clearing the VPOS flag while the INTEN.VPOSINTEN bit is 1.

GR1UFCLR Bit (GR1UF Flag Clear)

Writing 1 to this bit clears the STMON.GR1UF flag. This bit is only valid for the GR1UF flag, and does not directly affect the other status and interrupt request signals. However, a GR1UF interrupt request is negated by clearing the GR1UF flag while the INTEN.GR1UFINTEN bit is 1.

GR2UFCLR Bit (GR2UF Flag Clear)

Writing 1 to this bit clears the STMON.GR2UF flag. This bit is only valid for the GR2UF flag, and does not directly affect the other status and interrupt request signals. However, a GR2UF interrupt request is negated by clearing the GR2UF flag while the INTEN.GR2UFINTEN bit is 1.

51.2.58 Detected Status Monitor Register (STMON)

Address(es): GLCDC.STMON 000E 144Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	GR2UF	GR1UF	VPOS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VPOS	Specified Line Notification Detection Flag	0: Specified line notification has not detected 1: Specified line notification has detected	R
b1	GR1UF	Graphic 1 Underflow Detection Flag	0: No underflow detected in graphic 1 1: Underflow detected in graphic 1	R
b2	GR2UF	Graphic 2 Underflow Detection Flag	0: No underflow detected in graphic 2 1: Underflow detected in graphic 2	R
b31 to b3	—	Reserved	These bits are read as 0	R

VPOS Flag (Specified Line Notification Detection Flag)

This flag indicates that the specified line notification was detected in graphic 2. When this flag is 1, it indicates that the specified line notification was detected in graphic 2 at some time in the past. It does not necessarily mean that graphic 2 is currently processing the specified line. When this flag is 0, it indicates that no specified line notification was detected after the GLCDC operation was enabled. When the interrupt request signal is enabled and is cleared while this flag is 1, the corresponding interrupt request signal VPOS is negated, but this does not affect the status of graphic 2.

In addition, for the first frame after a software reset, the specified line notification signal is informed at the first scanline regardless of the value set in the GR2CLUTINT.LINE[10:0] bits. At this time, if the DTCTEN.VPOSDTC bit is set to 1, this flag is set to 1. Clear the flag without any processing required for the specified line notification. For detailed procedure, refer to section 51.3.1.1, Start of Operation.

GRnUF Flag (Graphic n Underflow Detection Flag) (n = 1, 2)

These flags indicate the detection of underflow of graphic n. When this flag is 1, it indicates that an underflow was detected in graphic n at some time in the past. It does not necessarily mean that graphic n is currently in the underflow status. When this flag is 0, it indicates that no underflow was detected after the GLCDC operation was enabled.

When setting the STCLR.GRnUFCLR bit to 1 while the INTEN.GRnUFINTEN bit is 1 and this flag is 1, the GRnUF interrupt request is negated, but this does not affect the status of graphic n.

Even if the current graphics data is not required (the GR1AB1.DISPSEL[1:0] bit are 00b (background color) or 01b (lower-layer graphics)), if the DTCTEN.GRnUFDTC bit is 1 (detection is enabled), this flag becomes 1 at the start of the graphics area. To avoid unnecessary detection flag settings or interrupt request signal assertions, when the current graphics data is not required, set the DTCTEN.GRnUFDTC bit and the INTEN.GRnUFINTEN bit to 0.

In addition, for the first frame after a software reset, the GRnMON.UFST flag may become 1 before the VS signal is asserted depending on the value set in the BGSYNC.VP[3:0] bits. At this time, if the DTCTEN.GRnUFDTC bit is 1, this flag becomes 1. Clear the flag without any processing for underflow. For detailed procedure, refer to section 51.3.1.1, Start of Operation.

51.2.59 Panel Clock Control Register (PANELCLK)

Address(es): GLCDC.PANELCLK 000E 1450h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	PIXSEL	—	—	—	CLKSEL	—	CLKEN						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	DCDR[5:0]	Clock Division Ratio Setting *1, *2	b5 b0 000000: Divide-by-2 000001: No division 000010: Divide-by-2 000011: Divide-by-3 000100: Divide-by-4 000101: Divide-by-5 000110: Divide-by-6 000111: Divide-by-7 001000: Divide-by-8 001001: Divide-by-9 001100: Divide-by-12 010000: Divide-by-16 011000: Divide-by-24 100000: Divide-by-32 Settings other than the above are prohibited.	R/W
b6	CLKEN	Panel Clock Output Enable	0: Disable LCD_CLK output 1: Enable LCD_CLK output	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b8	CLKSEL	Clock Source Select*1	0: Select LCD_EXTCLK (external clock) 1: Select PLL clock	R/W
b11 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	PIXSEL	Pixel Clock Select*1, *3	0: No frequency division (parallel RGB) 1: Divided-by-4 (serial RGB)	R/W
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b19 to b16	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b20	—	Reserved	This bit is read as 1. Writing to this bit has no effect.	R
b23 to b21	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R
b24	—	Reserved	This bit is read as 1. Writing to this bit has no effect.	R
b31 to b25	—	Reserved	These bits are read as 0. Writing to these bits have no effect.	R

Note: Do not rewrite this register while the BGMON.EN flag is 1 (in operation). Set the required settings before setting the BGEN.EN bit to 1 (enable operation). Operation is not guaranteed if the register is rewritten during operation.

Note 1. Rewrite these bits when the CLKEN bit is 0

Note 2. The frequency of the LCD_CLK should not exceed 30 MHz when these bits are set.

Note 3. Set the same value as the OUTSET.FRQSEL bit.

DCDR[5:0] Bits (Clock Division Ratio Setting)

These bits are used to set the division ratio of the LCD_CLK (panel clock). Set the bits after selecting the clock source by the CLKSEL bit. Operation is not guaranteed if a value not listed is set or if the frequency of the LCD_CLK exceeds 30 MHz.

CLKEN Bit (Panel Clock Output Enable)

This bit enables or disables the LCD_CLK (panel clock) output. When the LCD_CLK is enabled, the PXCLK (pixel clock) is generated according to the setting of the PIXSEL bit.

Set this bit to 0 before changing the PIXSEL bit, the CLKSEL bit, or the DCDR[5:0] bits. Operation is not guaranteed if any setting is changed while the LCD_CLK is being output.

CLKSEL Bit (Clock Source Select)

This bit selects the clock source of the LCD_CLK and PXCLK from either the LCD_EXTCLK or PLL clock.

PIXSEL Bit (Pixel Clock Select)

This bit selects the frequency of the PXCLK. For parallel RGB output, set this bit to 0 to set the frequency of the PXCLK to be the same as the LCD_CLK frequency (no frequency division). For serial RGB output, set this bit to 1 to set the frequency of the PXCLK to one quarter of the LCD_CLK frequency.

The setting of this bit must match the setting of the OUTSET.FRQSEL bit. Set this bit to the same value as that of the FRQSEL bit. Otherwise, operation is not guaranteed.

Figure 51.19 shows the structure of the pixel/panel clock generator.

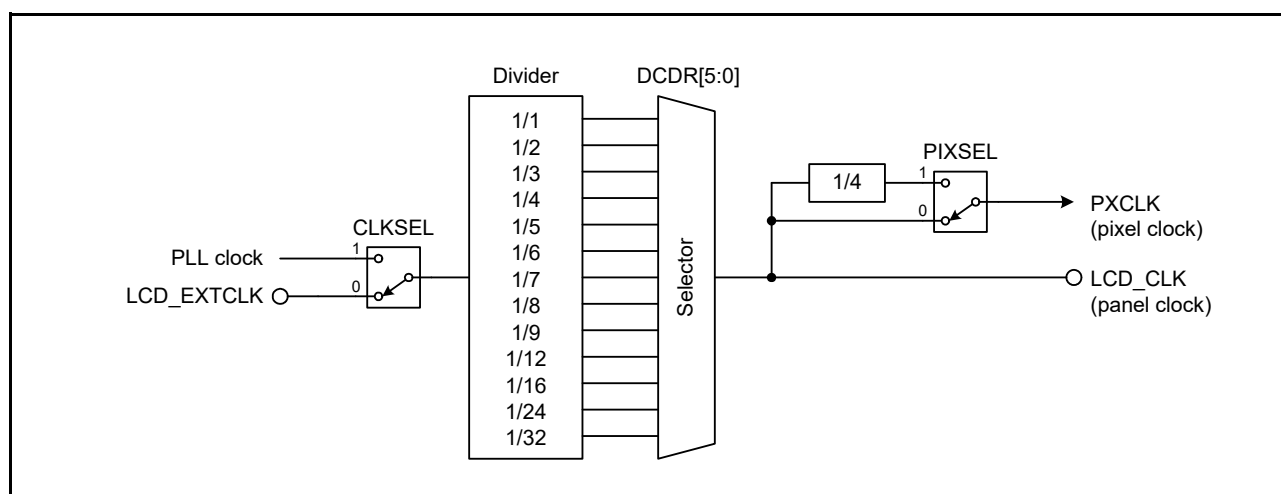


Figure 51.19 Structure of Clock Generator

51.3 Operation

51.3.1 Overall Controls

The GLCDC consists of six blocks and each block has the independent functions as shown in Figure 51.20. The vertical and horizontal synchronization signal (VS, HS, VE, and HE) and image data in the RGB(888) format shown in Figure 51.21 are conveyed between the four blocks where image data are processed.

The image data are processed in synchronization with the pixel clock (PXCLK). The LCD_CLK (panel clock) is synchronized with the PXCLK and the frequency of the LCD_CLK is the same as or (for serial RGB(888)) four times that of PXCLK.

The registers that control operations and specify the parameters are connected to the internal peripheral bus and operated in synchronization with PCLKA (peripheral bus clock). The circuits for reading the graphics data from internal main bus 2, access to the CLUT memory, and extending the graphics data to the ARGB(8888) format also run in synchronization with PCLKA.

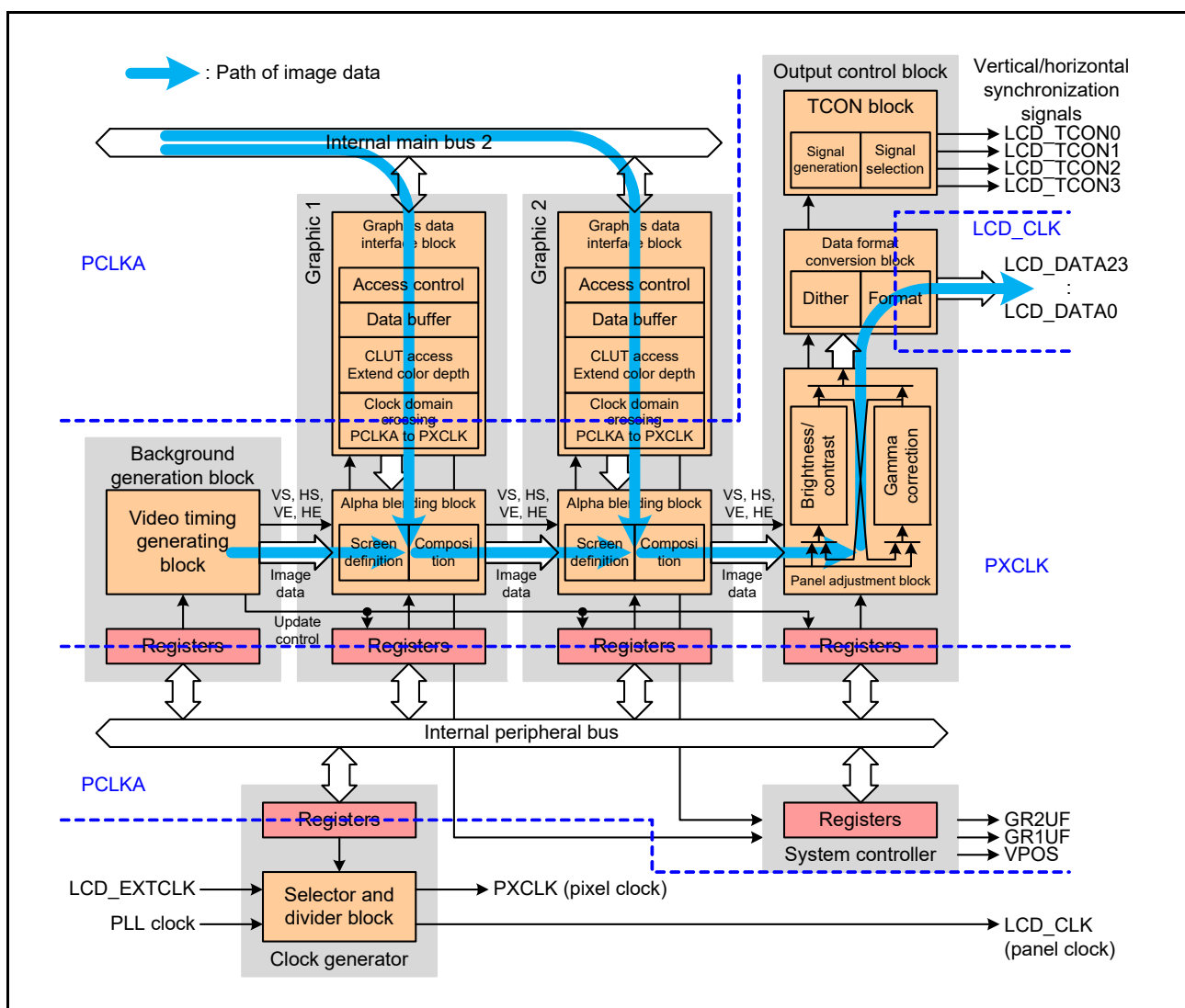


Figure 51.20 GLCDC Structure

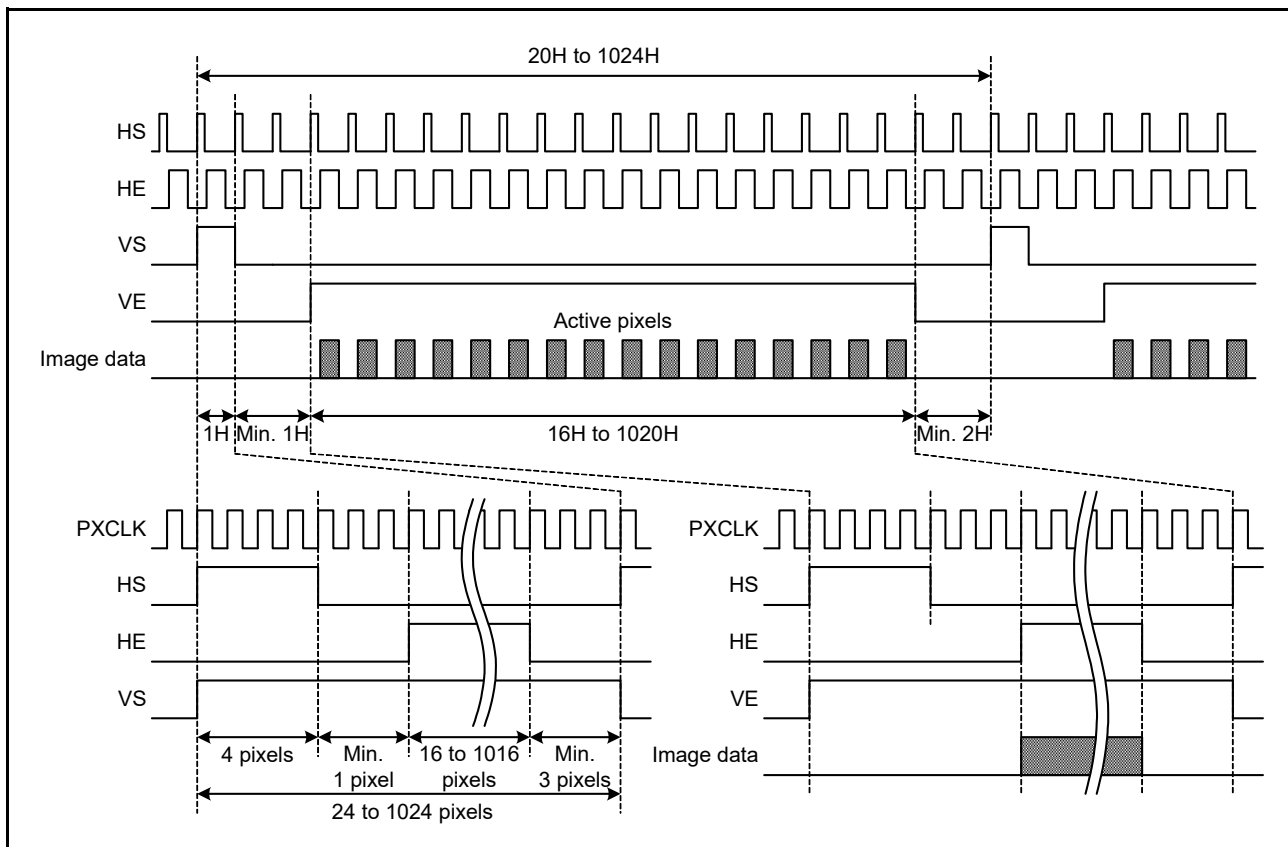


Figure 51.21 Image Data Carrier Signals in the GLCDC

The graphic 1, graphic 2, and output control blocks have no individual operation enable bits. When the software reset is released in the background generation block, these blocks wait for VS, HS, VE and HE signals, and for input of the image data (RGB(888)). Once the blocks detect the assertion of the VS signal, they start to operate in frame units according to the settings of the registers. For the system control block, releasing the software reset in the background generating block allows it to monitor the status of graphic 1 and 2 in accordance with the register settings and assert interrupt request signals. These status flags are intended for observing the internal status and do not affect the internal operations. The operation between start and stop of the GLCDC is outlined in this section.

51.3.1.1 Start of Operation

The procedure for starting the GLCDC is described as follows.

- (1) After releasing the GLCDC from the module stop state, set the BGEN.SWRST bit to 1 to release the GLCDC from a software reset.
- (2) Select the output data format. Set the frequency of the LCD_CLK and PXCLK to suit the format and set the PANELCLK.CLKEN bit to 1. If frequency settings which are not suitable for the output data format are attempted, operation is not guaranteed. Wait until PXCLK and LCD_CLK are supplied and the BGMON.SWRST flag becomes 1.
- (3) Set values in the registers that are required for operations. No particular rules apply to the order of setting of the registers. However, do not set the BGEN.EN, BGEN.VEN, GRnVEN.VEN, GAMxVEN.VEN, OUTVEN.VEN, and INTEN.GRnUFINTEN bits to 1 ($n = 1, 2$; $x = G, B, R$).
 - When displaying the graphics data on the frame buffer by graphic 1 and 2, set 1 to the GRnFLMRD.RENB bit (graphics data read enable).

- When using the index color, write the color data to the CLUT memory (the GRnCLUTm[k] registers (n = 1, 2; m = 0, 1; k = 0 to 255)). ARGB(1555) uses registers GRnCLUTm[0] and GRnCLUTm[128], CLUT(1) uses registers GRnCLUTm[0] and GRnCLUTm[1], CLUT(4) uses registers from GRnCLUTm[0] to GRnCLUTm[15], and CLUT(8) uses registers from GRnCLUTm[0] to GRnCLUTm[255].
 - Set the DTCTEN.VPOSDTC bit to 1 (enable detection of specified line notification). When an interrupt is to be used, also set the INTEN.VPOSINTEN bit to 1.
- (4) Set the BGEN.VEN bit and the BGEN.EN bit to 1 at the same time. This setting allows output of the signals VS, HS, VE, and HE, and the image data (RGB(888)) from the background generating block. The image data is that of the BGCOLOR register in the active pixel area (in which both VE and HE are 1) and 000000h (black) in the blanking interval.
 - (5) The STMON.VPOS flag becomes 1 at the first scanline of the first frame. Confirm that the flag is 1 and then write 0000 0007h in the STCLR register to clear all flags. After that, re-configure the DTCTEN and INTEN registers according to your application.
 - (6) Each block detects the assertion of the VS signal output from its previous stage (background generating block for graphic 1, graphic 1 for graphic 2, and graphic 2 for the output control block) and starts operation in accordance with its register settings. Each block completes its operation frame by frame. When the assertion of the VS signal is detected, each block recognizes it as start of frame and initializes the status. When the VEN bit is 1, the value of the register is reflected to the internal operations.

51.3.1.2 Rewriting Registers During Operations

The GLCDC has registers, which values can be updated and reflected in internal operations without stopping the operation of individual blocks. Setting the VEN bit to 1 in the block to be updated leads to a reflection of the value of an updated register to the internal operations immediately after assertion of the VS signal.

However, if the contents of the CLUT memory in use (selected by the GRnCLUTINT.SEL bit) are changed, the value will be reflected to the internal operations immediately after the change without waiting for assertion of the VS signal. To update the value of the CLUT memory in synchronization with the VS signal, rewrite the contents of the CLUT memory which is not in use, and then switch the CLUT memory to be used by using the GRnCLUTINT.SEL bit. After that, set the BGEN.VEN bit or GRnVEN.VEN bit to 1.

The procedure for changing parameters during operations is as follows.

- (1) Confirm that the VEN bit in the target block is 0. Operation is not guaranteed if the register is modified while the VEN bit in the target block or the BGEN.VEN bit is 1.
- (2) Rewrite the value of the target registers.
- (3) To reflect the value of the register only in a target block, set the VEN bit of the block to 1. When the register values in multiple blocks are to be reflected, set the BGEN.VEN bit to 1.
- (4) Confirm that the VEN bit that was changed to 1 has returned to 0. Once the value of this bit is 0, the value of the target register will have been reflected in internal operations. If this bit is still 1, then the change may not have been reflected. In addition, the VEN bit in each block is cleared to 0 once the value of the target register is internally reflected, but the BGEN.VEN bit will not be cleared to 0 until the VE signal is negated.

Figure 51.22 shows the operation of the signals output by the background generating block and the status flags.

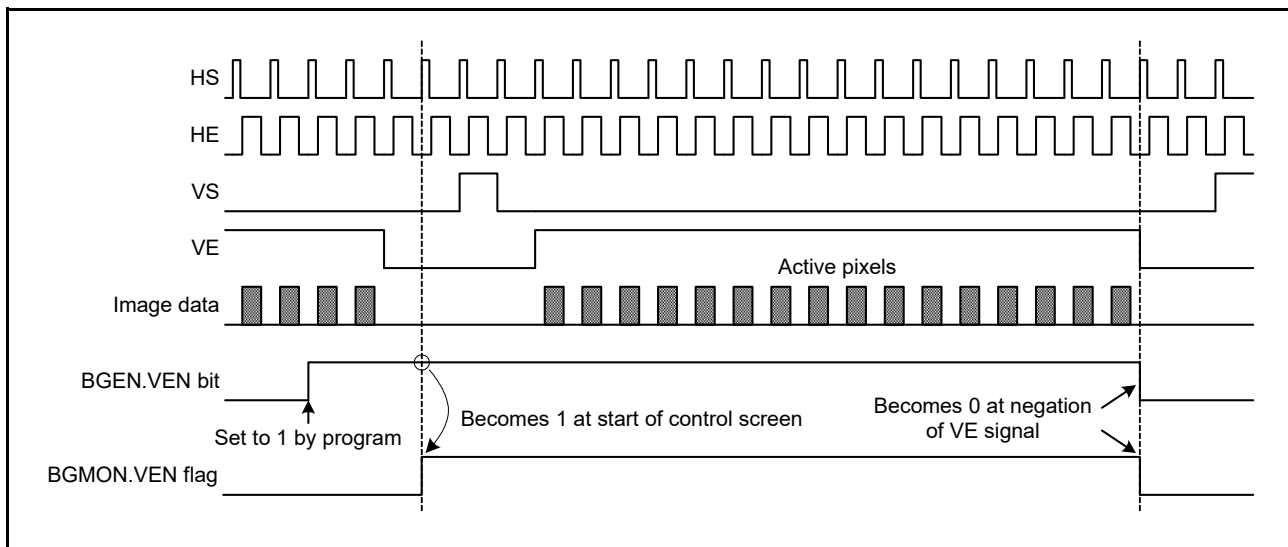


Figure 51.22 Control Signals for Register Value Reflection to Internal Operations

51.3.1.3 Stopping and restarting in normal operation

The procedure for stopping and restarting the GLCDC is as follows.

- (1) Confirm that the VEN bit of each block is set to 0.
- (2) Set the BGEN.EN bit to 0.
- (3) Confirm that the BGMON.EN flag has been set to 0. Operation of the GLCDC has not stopped until the value of this flag becomes 0. In addition, the EN flag is set to 0 at the end of the frame in the background generating block, not of that in the signals output to the LCD panel. When completion of operation of all blocks is needed, wait for a certain period (e.g. time equivalent to a single horizontal line) after the EN flag has been set to 0. In general, if the BGEN.SWRST bit is set to 0 (software reset) immediately after confirming that the BGMON.EN flag has been set to 0, there should not be any problems due to the vertical blanking interval.
- (4) Set the BGEN.SWRST bit to 0 to apply a software reset to the GLCDC. Registers other than the CLUT memory are initialized by a software reset.
- (5) Start the GLCDC in accordance with section 51.3.1.1, Start of Operation.

When restarting the GLCDC after rewriting to a register without a software reset, wait till the end of the frame in the output signals to the LCD panel before starting the GLCDC. There is no problem for the GLCDC itself to start operation in response to the assertion of the VS signal. However, if the vertical blanking interval or a single horizontal line is short, this may affect the operation of the LCD panel. For details, check the specifications of the LCD panel.

Figure 51.23 shows the changes in signal lines and flags for stopping and restarting normal operation. When access to the frame buffer has not been completed by the frame end because an underflow occurs in graphic 1 or 2, or register settings are inadequate for access to the graphics data, the GLCDC enters the state of an abnormal stop even if the BGEN.EN bit is set to 0 to stop the background generating block.

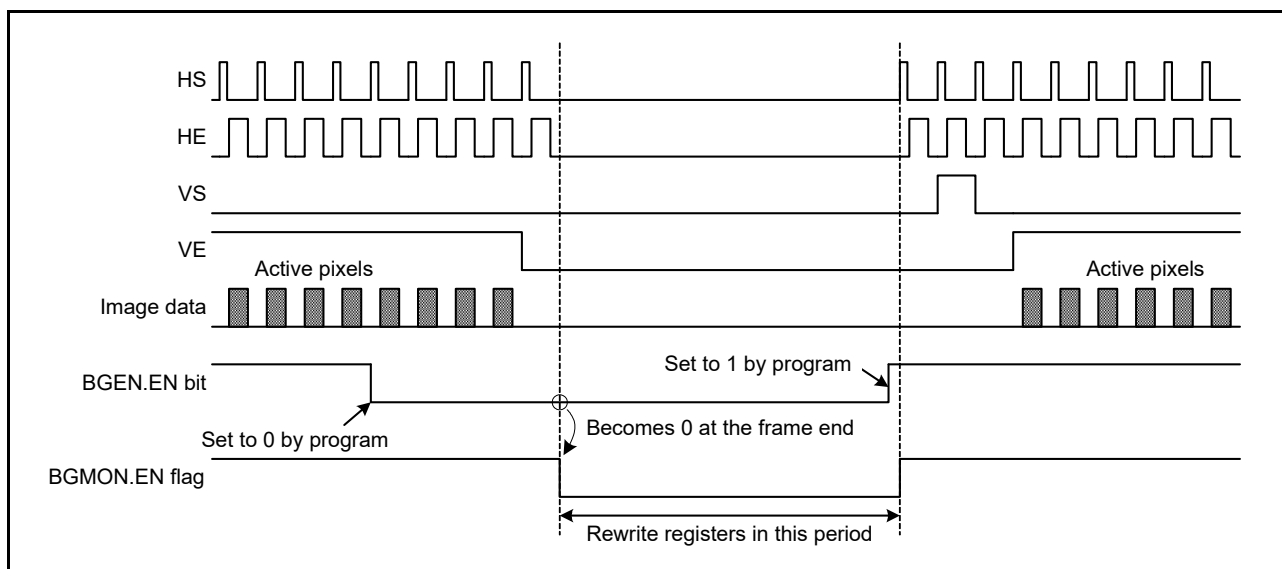


Figure 51.23 Stop and Restart in Normal Operation

51.3.1.4 Restarting in abnormal operation

The GLCDC stops operation due to abnormality in the following cases.

- When the BGEN.SWRST bit is set to 0 (software reset) or a hardware reset is applied while the BGEN.EN bit or BGMON.EN flag is 1 (in operation).
- When an unintended access to the frame buffer is made even after the BGEN.EN bit is set to 0 and the BGMON.EN flag has become 0.

Either of the cases, the internal state of the GLCDC has been initialized and the registers can normally be accessed (except for hardware resets). However, in the case of (b), access to the frame buffer may not finish. Therefore, even after setting the BGEN.EN bit to 0 and confirming that the BGMON.EN flag has become 0, set the BGEN.SWRST bit to 0 (software reset). Afterwards, start the GLCDC in accordance with section 51.3.1.1, Start of Operation.

51.3.2 Screen Definition

The reference signals for GLCDC operations are generated by the background generating block, and the graphic 1 and 2 blocks and output control block operate based on the sequentially propagated VS (vertical synchronization) signal and HS (horizontal synchronization) signal, VE (vertical data enable) signal and HE (horizontal data enable) signal.

Figure 51.24 shows the definition of the screen. The reference point (frame start point) of the background screen cannot be determined by the output signals. The point shown in the figure is virtual and provided only for register settings. The reference points of the graphic 1 and 2 blocks and output control block are the assertion of the VS signal, which is input from the previous block.

The active display area or rectangular alpha blending area are specified by setting the position and width from this reference point for graphic 1 and 2 blocks. In the panel adjustment block, adjustment is performed by using the VE and HE signals for the entire active display area which is output from graphic 2. The control signals to be output to the LCD panel are generated in the TCON block based on the VS and HS signals.

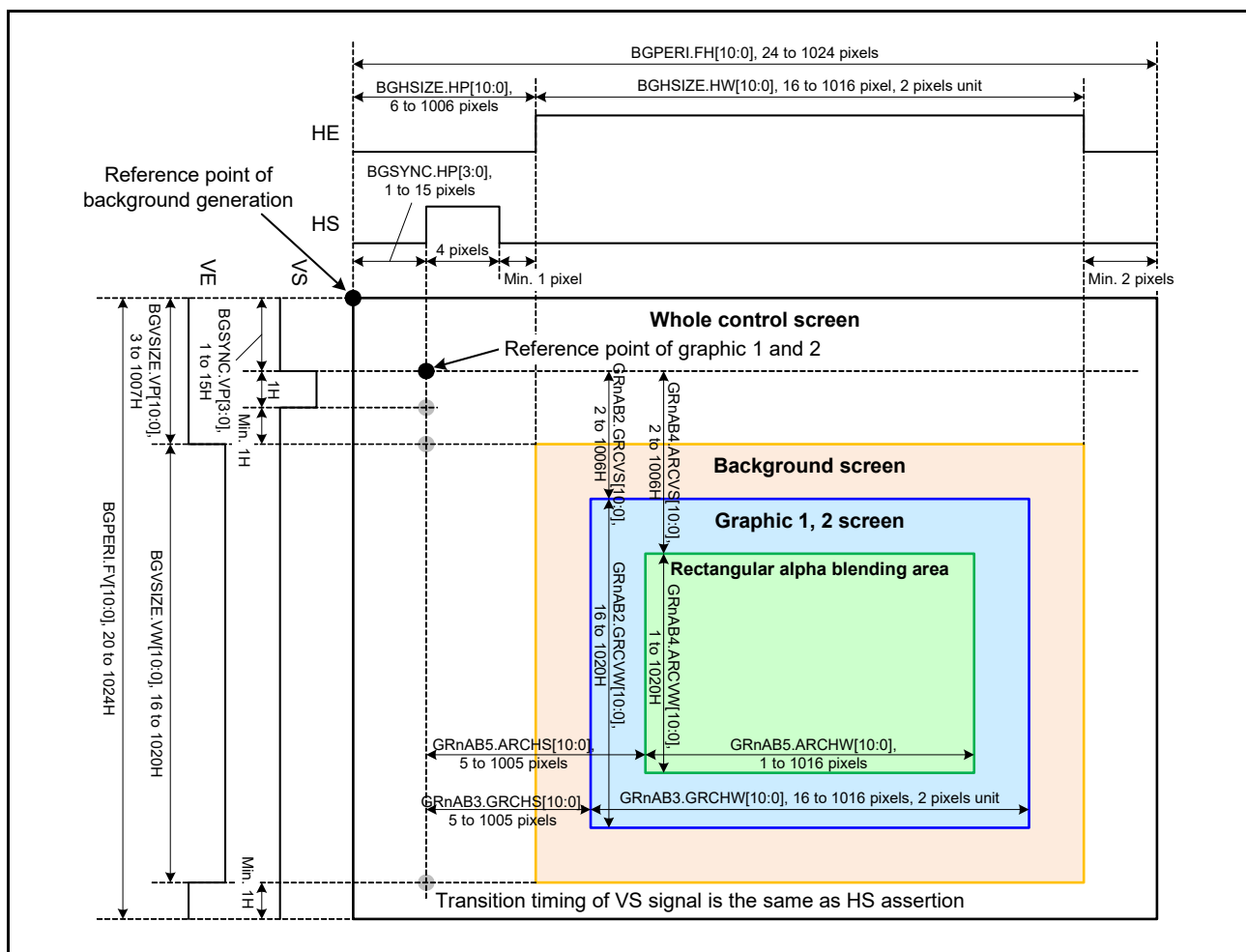


Figure 51.24 Definition of Screen

51.3.3 Status Detection and Interrupts

The GLCDC can detect the three types of status described in this section. To detect each status, set the corresponding bit in the DTCTEN register to 1.

51.3.3.1 Graphic n Underflow Detection (n = 1, 2)

This function is equipped with both graphic 1 and 2.

When the graphics data interface block fails to supply required graphic data to the alpha blending block (the valid data was not stored in the 4-stage ring buffer), the STMON.GRnUF is set to 1. The underflow status in the graphic n is cleared when the VS (vertical synchronization) signal is asserted, but the STMON.GRnUF flag is not cleared until the STCLR.GRnUFCLR bit is set to 1.

51.3.3.2 Specified Line Detection

This function detects that the target of data to be processed by graphic 2 is the line specified in the GR2CLUTINT.LINE[10:0] bits. In fact, the detection by using the VS and HS signals is applied regardless of whether graphic 2 operations are actually to proceed. Detection proceeds on the assertion of the HS (horizontal synchronization) signal, but not at the start of the processing of active pixels. Every time the HS signal for the specified line is asserted, the STMON.VPOS flag is set to 1. To clear the STMON.VPOS flag, the STCLR.VPOSCLR bit must be set to 1.

51.3.3.3 Interrupts

The GLCDC has three interrupt request output signals (GR2UF, GR1UF, and VPOS) that correspond to the three states above. Setting the corresponding bit in the INTEN register to 1 leads to the output of the interrupt request signal on detection of the corresponding state. When the corresponding bit in the STCLR register is set to 1 and the corresponding flag in the STMON register is cleared, or the corresponding bit in the INTEN register is set to 0 (disable interrupts) while the interrupt request output signal is being output, the interrupt request signal can be negated.

The interrupt request signal generating circuit is configured as shown in Figure 51.25.

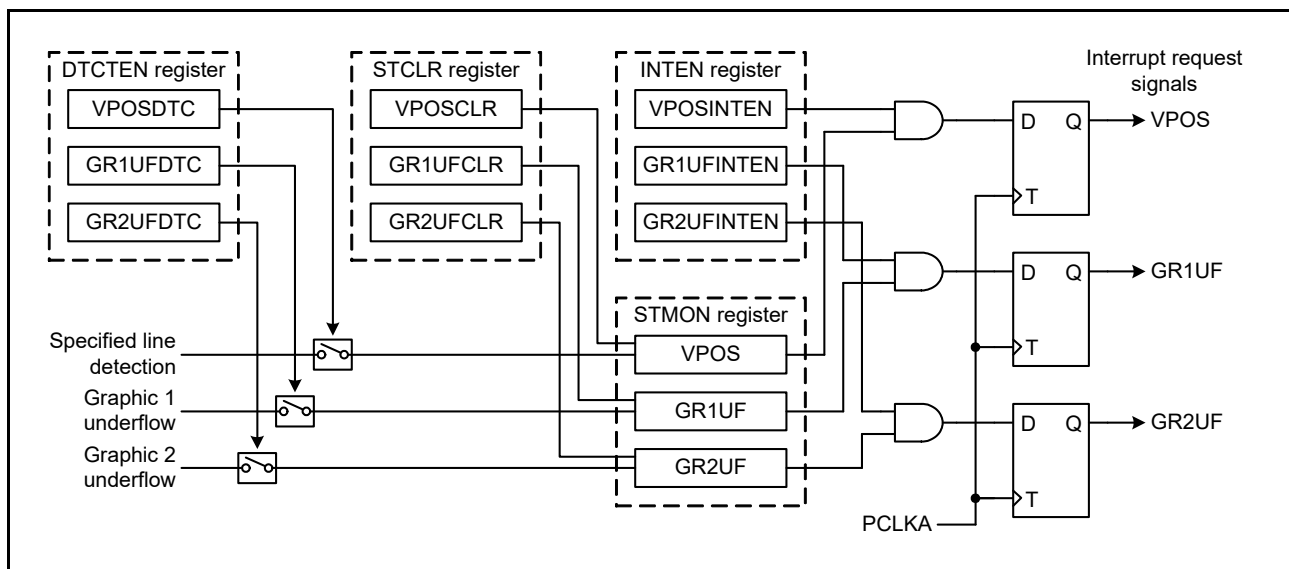


Figure 51.25 Interrupt Request Signal Generating Circuit

51.4 Usage Notes

51.4.1 Setting the Priority Order of the Bus Masters

The GLCDC acquires graphics data from the frame buffer through internal main bus 2. The graphic 1 (GLCDC-GRA1) and graphic 2 (GLCDC-GRA2) blocks in the GLCDC, the texture and RLE interface (DRW2D-TX) and display-list and frame-buffer interface (DRW2D-FB) in the DRW2D, the EDMAC, DMAC, and DTC can operate as bus masters of the internal main bus.

After a reset, the priority order of these bus masters is as follows:

GLCDC-GRA1 > DRW2D-TX > DRW2D-FB > GLCDC-GRA2 > EDMAC > DMAC > DTC.

In some cases, using the bus masters with this order of priority may lead to underflows of data for graphic 2. Change the priority order of the bus masters as required.

For details, refer to section 16, Buses, or section 18, DMA Controller (DMACa).